

**Determinants of Infant Mortality in India:
A Survival Analysis Perspective Using NFHS-5
Data (2019-21)**

Submitted by
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Chapter 1

INTRODUCTION

The term infant mortality which is said to be the death of a live-born child before the completion the first year of its life which stands as one of the most powerful indicator of the society's health and developmental conditions. The infant mortality rate highlights mainly the biological vulnerability and the factors like maternal health, nutritional adequacy, environmental sanitation, healthcare infrastructure and the depth of socioeconomic inequality which totally shapes the early months of a child's life. The rise in IMR reveals not only a deficit in pediatric care but also a broader failure our development systems to protect the most vulnerable members of our society. On the other hand the sustained reduction of the infant mortality has always served as one of the most reliable markers of the human progress.

India's position in this global narrative brings both hope and reality check about the situations regarding the existential development. The NFHS-5 data reveal that India continues to carry a disproportionate share of the global infant death burden and that around four-fifths of all under-five deaths in the country still occur in infancy (IIPS & ICF, 2021; UNICEF, 2021). What makes the situation analytically compelling is not only the national average but also the huge variations concealed within where the states such as Kerala and Tamil Nadu have achieved infant mortality rates that approach the performance of upper-middle-income European countries, while the states within the central and eastern belt from Uttar Pradesh, Bihar, Madhya Pradesh and Rajasthan continue to record rates that would not be misplaced in sub-Saharan Africa (Ram et al., 2013). Rural children face mortality risks roughly 1.5 times those of their urban counterparts; children born to mothers with no schooling face more than twice the risk of those born to highly educated mothers; and children in the poorest wealth quintile die at nearly twice the rate of those in the wealthiest (IIPS & ICF, 2021). These gradients are not only statistical facts but also the demographic image of the structural inequality and they carry a precise policy implication by reducing the infant mortality in the twenty-first century India which is not simply a matter of implementing more medical technology but also due to the understanding the determinants of survival at each levels of the child, maternal, and socioeconomic constraints.

This dissertation mainly addresses this challenge by applying a set of the survival analysis techniques to the NFHS-5 Birth Recode file, covering about 232,159 live births recorded across all states and union territories of India. Survival analysis is not merely a method-

ological preference for this research but also a statistical necessity. Infant mortality data are time-to-event data which depicts that a child who dies at eleven months carries fundamentally different epidemiological information from one who dies on the first day of life, but yet the logistic regression that codes both as “dead” treats them identically, discarding the time dimension entirely. In addition to that a great majority of births in the dataset which is about 96.5 per cent are right-censored which means that the child was still alive at the time of the interview. The standard regression approaches that discard censored observations or combine them with uncensored survivors produce estimates that are biased and inconsistent (Kleinbaum & Klein, 2012). Survival analysis was developed precisely to handle both of these features correctly, extracting the maximum statistical information from the data while respecting the structural reality that most infants survive.

This study is theoretically followed by the analytical framework of Mosley & Chen (1984), which provides a principled distinction between the socioeconomic determinants of child mortality factors such as maternal education, household wealth and place of residence and the biological determinants through which they must necessarily operate including the maternal factors (birth order, birth spacing, age at first birth), nutritional adequacy, environmental contamination, injury, and personal illness control. This hierarchical logic motivates the three-level covariate structure implemented throughout the analysis.

The reduction of infant and child mortality has been a central objective of the international development agenda since the Declaration of Alma-Ata in 1978. The Millennium Development Goals provided this agenda with explicit numerical targets between 2000 and 2015 with a goal for calling for a two-thirds reduction in under-five mortality relative to the 1990 baseline.

Progress toward these targets has been real but uneven. Globally, the under-five mortality rate fell from 93 per 1,000 in 1990 to 38 per 1,000 in 2019, representing the saving of approximately 5.2 million children’s lives per year (WHO, 2022). That said the aggregate masks a profound inequality where a child born in the least-developed countries remains roughly fifty times more likely to die before age five than one born in a high-income country, and within low- and middle-income countries, the gradient across wealth and education groups is steep and persistent (Chalasani, 2012; Mosley & Chen, 1984).

India’s mortality transition since independence has been extensively studied. An important early contribution by Bhat (2002) offered a critical reassessment of the pace of mortality decline in northern India, arguing that regional data quality problems meant that official statistics consistently overestimated the pace of improvement in the Hindi-belt states. His work was an early reminder that national averages in a country of India’s size and diversity are statistical artifacts that can be uncertain as much as they reveal.

The trend across five rounds of the National Family Health Survey charts both the achieve-

ment and the distance that remains. Table 1.1 traces the evolution of key mortality indicators from NFHS-1 to NFHS-5.

Table 1.1: Trends in Neonatal, Infant, and Under-Five Mortality Rates in India, NFHS-1 to NFHS-5 (deaths per 1,000 live births)

Survey Round	Period	NMR	IMR	U5MR
NFHS-1	1992–93	52	79	109
NFHS-2	1998–99	45	68	95
NFHS-3	2005–06	39	57	74
NFHS-4	2015–16	30	41	50
NFHS-5	2019–21	25	35	42

Source: IIPS & ICF (2021).

The data in Table 1.1 document a cumulative reduction of 56 per cent in the infant mortality rate between 1992–93 and 2019–21, one of the more significant public health achievements of the post-reform era. Nevertheless, the NFHS-5 neonatal mortality rate of 25 per 1,000 live births remains well above the SDG 3.2 target of 12, and the under-five mortality rate of 42 per 1,000 still exceeds the SDG target of 25. Four-fifths of all under-five deaths continue to occur during infancy (IIPS & ICF, 2021), underscoring the importance of the infant period as the focus of this analysis.

Perhaps the most consistent finding across the Indian mortality literature is that the place of birth remains a powerful determinant of survival outcomes, independent of individual and household characteristics. Ram et al. (2013) analysed neonatal and child mortality across 597 districts of India between 2001 and 2012, documenting that high-mortality clusters were geographically concentrated in Uttar Pradesh, Bihar, Madhya Pradesh, and Rajasthan, which accounted for a disproportionate share of infant deaths even as their rates fell over time. Singh & Pathak (2014) confirmed the presence of spatial clustering of mortality risk, with high-mortality districts surrounded by other high-mortality districts in patterns that persisted across two decades of surveys.

1.1 Major Government Initiatives and Their Relevance

The improvements in infant mortality documented in Table 1.1 reflect not only natural demographic transition but also deliberate policy interventions. The *Janani Suraksha Yojana* (JSY), launched in 2005 under the National Rural Health Mission, provides conditional cash transfers to incentivise institutional delivery among poor and low-parity women. NFHS-5 recorded a dramatic increase in institutional delivery rates, from 79 per cent in 2015–16 to 89 per cent in 2019–21.

The *National Health Mission* (NHM) has expanded antenatal care coverage, skilled birth attendance, and postnatal follow-up across the country, with particular emphasis on rural and

tribal areas. The present study's central finding that adequate antenatal care reduces infant mortality hazard by approximately 70 per cent directly quantifies the mortality dividend attributable to sustained investment in this programme. The *Poshan Abhiyan* (National Nutrition Mission), launched in 2018, targets interlinked problems of child and maternal undernutrition that remain among the most powerful proximate determinants of infant death. Despite these achievements, the NFHS-5 data reveal persistent gaps: only 58.4 per cent of mothers attended the WHO-minimum four or more antenatal visits, and 11 per cent of births still occurred outside health facilities entirely (IIPS & ICF, 2021).

1.2 Rationale for a Survival Analysis Approach

The methodological literature on infant mortality has been dominated, until relatively recently, by logistic regression of a binary death indicator and linear regression of grouped mortality rates. While these approaches are computationally tractable and widely understood, they suffer from two structural limitations that are particularly in the context of birth-history data from large-scale household surveys.

The first is the time-to-event problem. A child who dies at eleven months has lived almost the entire period of biological vulnerability that defines infancy. Infectious diseases that claim post-neonatal lives are largely preventable through immunization, safe water, and nutritional support, and the policy levers that would have saved this child differ fundamentally in kind from those relevant to a child who dies from birth asphyxia on day one. A logistic regression that codes both outcomes as “dead” discards this epidemiologically essential information.

The second limitation is the censoring problem. In the present dataset, 96.5 per cent of the 232,159 live births are right-censored: the child was alive at the time of the interview, but the true failure time is unknown and bounded only from below by the child's age at survey. An ordinary regression that either discards these observations or treats them as equivalent to children observed to survive will in either case produce biased and inconsistent estimates (Cox, 1972; Kaplan & Meier, 1958).

Survival analysis was developed specifically to address both of these problems. The Kaplan–Meier estimator (Kaplan & Meier, 1958) provides a non-parametric estimate of the survival function that correctly incorporates right-censored observations as known lower bounds on survival time. The Cox proportional hazards model (Cox, 1972) estimates covariate effects on the instantaneous hazard of death as a continuous function of time, using only the rank ordering of failure times and the partial likelihood function. Neither method discards censored observations; both extract the full information content of the data. Their application to birth-history data from DHS surveys is methodologically well established in the demographic literature (Roy & Unisa, 2000; Singh & Tripathi, 2013; Kleinbaum & Klein, 2012).

Kaplan–Meier estimation for non-parametric group comparisons; standard Cox proportional hazards regression for estimation of adjusted hazard ratios in the full analytical sample (Model 1 with birth order) and a non-first birth sub-sample (Model 2 with birth interval); and a stratified Cox model using primary sampling unit cluster identifiers as strata to account for unobserved heterogeneity generated by the NFHS-5 cluster sampling design.

1.3 The National Family Health Survey, Round 5

1.3.1 Overview

The primary data for the present study were drawn from the fifth round of India’s National Family Health Survey, hereafter NFHS-5, conducted during 2019–21 (IIPS & ICF, 2021). The NFHS is India’s principal nationally representative household survey instrument for health and demographic data, analogous to the Demographic and Health Survey (DHS) programme that operates across more than ninety low- and middle-income countries. The survey is implemented by the International Institute for Population Sciences (IIPS), Mumbai, under the Ministry of Health and Family Welfare, Government of India, with technical support from ICF International.

NFHS-5 is the most methodologically refined round of the series to date, and the first to provide district-level estimates for all 707 districts of India, necessitating a substantially expanded sample relative to earlier rounds.

1.4 Theoretical Framework: Mosley-Chen

Empirical analyses of child mortality are most defensible when grounded in a theoretical model that imposes a principled ordering on the determinants under study. The present study is organised around the analytical framework of Mosley & Chen (1984), which remains the most widely cited and methodologically perfect for understanding child survival in low-income settings.

Mosley & Chen (1984) drew a fundamental distinction between *socioeconomic determinants* and *proximate determinants*. In their model there is no socioeconomic variable can directly produce the biological outcome of death; it can only do so by activating one or more of five proximate pathways: maternal factors (age at birth, birth order, birth spacing), nutritional adequacy, environmental contamination, injury, and personal illness control.

The practical consequence for the present study is a three-tier covariate structure:

Tier 1 - Child-level proximate determinants:

Sex of the child, birth order, preceding birth interval, and twin status. These variables directly reflect the biological conditions of the birth and the immediate post-natal environment.

Tier 2 - Maternal intermediate determinants:

Maternal educational attainment, age at first birth, antenatal care utilization, and place of delivery. These variables shape the quality of the obstetric and post-natal environment through the mother's knowledge, health-seeking behaviour, and access to skilled care.

Tier 3 - Socioeconomic and contextual distal determinants:

Household wealth index, type of place of residence, and religious affiliation. These variables represent the structural conditions through which the proximate determinants are configured.

1.5 Dissertation overview

The rest of the study is organised as following. the Chapter 2 sets out the specific objectives of the study, Chapter 3 dives into the empirical literature on the infant mortality and their determinants, Chapter 4 focuses on the methodological architecture of the study in full detail, Chapter 5 provides the complete findings and the Chapter 6 includes the substantive discussion of the findings, and the Chapter 7 concludes the dissertation.

Chapter 2

OBJECTIVES OF THE STUDY

*Determinants of Infant Mortality in India:
A Survival Analysis Perspective Using NFHS-5 Data*

Despite significant declines in infant mortality over the past three decades, India continues to carry a disproportionate share of the global infant death burden. The fifth round of the National Family Health Survey (NFHS-5, 2019–21) recorded a national infant mortality rate of 35 deaths per 1,000 live births, yet this aggregate conceals vast within-country variation across states, districts, wealth groups, and religious communities. The persistence of this inequality, and the need for methodologically rigorous analysis of its determinants, motivates the present dissertation.

The study is grounded in the analytical framework proposed by Mosley & Chen (1984), which organises the determinants of child mortality into three hierarchical tiers: child-level proximate determinants (sex, birth order, preceding birth interval, twin status); maternal intermediate determinants (educational attainment, antenatal care utilisation, place of delivery, age at first birth); and socioeconomic and contextual distal determinants (household wealth, place of residence, religious affiliation). The empirical analysis applies a sequential suite of survival analysis methods — Kaplan–Meier estimation, Cox proportional hazards regression, and stratified Cox modelling — to the NFHS-5 Birth Recode file encompassing 232,159 live births and 8,095 infant deaths.

Specific Objectives

To identify and quantify the proximate, intermediate, and distal determinants of infant mortality in India using survival analysis applied to nationally representative birth-history data from the fifth National Family Health Survey (NFHS-5, 2019–21), and to distinguish individual-level causal mechanisms from geographic clustering effects through a stratified Cox model specification.

Specific Objectives

1. **To examine bivariate survival differentials** across all eleven determinant variables through Kaplan–Meier survival curves and log-rank, Breslow, and Tarone–Ware equality tests, stratified by child-level, maternal, and socioeconomic characteristics, using the NFHS-5 Birth Recode file ($N = 232,159$ live births; 8,095 infant deaths).
2. **To estimate the independent net effect of birth order** on infant mortality hazard through Cox Model 1, fitted on the full analytical sample ($N = 232,159$) with simultaneous adjustment for all eleven covariates across the three Mosley–Chen tiers, and to interpret the resulting hazard ratios in the context of the biological depletion hypothesis.
3. **To estimate the independent net effect of preceding birth interval** on infant mortality hazard through Cox Model 2, fitted on the non-first birth sub-sample ($N = 143,020$), and to determine whether the mortality risk associated with short intervals (< 24 months) is concentrated in the neonatal or post-neonatal phase using the Breslow test as a diagnostic.
4. **To account for unobserved cluster-level heterogeneity** introduced by the NFHS-5 stratified two-stage cluster sampling design through a Stratified Cox model incorporating primary sampling unit (PSU; V021) as a stratification variable, and to assess whether individual-level covariate effects are attenuated or amplified after cluster control.
5. **To distinguish robust individual-level determinants from geographic confounders** through cross-model comparison of hazard ratios across all three Cox specifications, classifying each covariate effect as robust (stable across all models), geographic (disappears after cluster stratification), or null (consistently non-significant).
6. **To derive evidence-based policy implications** for reducing infant mortality in India, with specific focus on the most tractable and highest-impact leverage points identified by the empirical analysis, and to situate these findings within the contemporary policy context of the National Health Mission, Janani Suraksha Yojana, and Ayushman Bharat.

Scope and Analytical Design at a Glance

Dimension	Details
Data source	NFHS-5 Birth Recode file (IABR7AFL.SAV), 2019–21
Analytical sample	232,159 live births; 8,095 infant deaths (IMR = 34.9 per 1,000)
Outcome variable	Time to infant death or right-censoring at 12 completed months
Covariates	11 variables across 3 Mosley–Chen tiers
Methods	Kaplan–Meier; Cox PH (Models 1 & 2); Stratified Cox
Software	IBM SPSS Statistics, Version 21
Geographic scope	All 28 states and 8 union territories of India
Reference period	Five years preceding the 2019–21 interview date

Guiding Hypotheses

The present study is guided by the following testable hypotheses, derived from the Mosley–Chen framework and the prior literature:

- H_1 : Female infants exhibit a lower hazard of death than male infants, reflecting the well-documented biological female survival advantage, which holds even after controlling for socioeconomic covariates.
- H_2 : The hazard of infant death increases monotonically with birth order beyond the second–third birth category, consistent with the maternal resource depletion hypothesis.
- H_3 : Short preceding birth intervals (< 24 months) are associated with significantly elevated infant mortality hazard relative to medium intervals (24–47 months), with the effect concentrated in the post-neonatal rather than neonatal period.
- H_4 : Antenatal care utilisation meeting the WHO minimum of four or more visits is the single most powerful individual-level protective factor for infant survival in India, retaining large and statistically significant protective effects after full covariate adjustment.
- H_5 : The apparent associations of household wealth and religious affiliation with infant mortality are substantially artefactual, reflecting geographic clustering of socioeconomic and religious groups in different mortality environments rather than individual-level causal pathways, and will attenuate markedly after PSU stratification.

H_6 : Maternal education has no independent effect on infant mortality hazard once household wealth is controlled, consistent with a view of education as a mediator of the wealth effect rather than an independent causal pathway.

Chapter 3

REVIEW OF LITERATURE

Infant mortality remains one of the most telling indicators of a nation's health and developmental trajectory. This chapter examines the substantive literature on the determinants of infant and child mortality in India, with particular attention to factors operating at three interconnected analytical levels such as the characteristics of the child, attributes of the mother, and broader socioeconomic and contextual conditions. Focusing on theoretical frameworks and empirical work across multiple rounds of the National Family Health Survey, the review builds the conceptual foundation for the survival analysis undertaken in the chapters that follow.

3.1 Global and National Epidemiological Context

3.1.1 The Global Picture

Advances in sanitation, nutrition, vaccination, and obstetric care drove infant mortality rates in high-income countries from well above 100 per 1,000 live births at the turn of the last century to single digits by its end (WHO, 2022). The Millennium Development Goals gave this momentum fresh impetus in the developing world, and the Sustainable Development Goals have continued that effort with specific targets: SDG 3.2 calls for reducing neonatal mortality to no more than 12 per 1,000 live births and under-five mortality to no more than 25 per 1,000 live births in every country by 2030 (UNICEF, 2021).

Yet the aggregate progress masks persistent inequalities. The risk of dying in the first year of life remains roughly fifty times higher for a child born in the least-developed countries than for one born in a high-income country (WHO, 2022). Even within low- and middle-income countries, the gradient between rural and urban areas, between the poorest and the wealthiest households, and between the most and least educated mothers is steep, well documented, and has narrowed only modestly in relative terms even as absolute mortality rates have fallen (Chalasani, 2012).

3.1.2 Trends in India

India's mortality transition has been extensively chronicled. Bhat (2002) offered an important early reassessment, arguing that while official statistics pointed to substantial improvement, regional data quality problems meant that declines in the northern states were systematically

overestimated. His work was an early and prescient reminder that national averages in a country of India's size and internal diversity are statistical artifacts that can obscure as much as they could reveal.

Subsequent research confirmed both the reality of decline and the persistence of inequality. Ram et al. (2013) used data from multiple rounds of nationally representative surveys to estimate mortality rates at the district level for the period 2001 to 2012. Their analysis documented that the neonatal mortality rate fell from approximately 44 per 1,000 live births in 2001 to 29 per 1,000 in 2012, yet districts in Uttar Pradesh, Bihar, Madhya Pradesh, and Rajasthan accounted for a disproportionate share of all infant deaths even as their rates fell a geographic concentration of disadvantage that remains a defining feature of Indian infant mortality in the NFHS-5 era.

3.1.3 Regional Heterogeneity and the Geography of Child Death

Perhaps the most consistent finding across the Indian mortality literature is that where a child is born matters enormously. Singh & Pathak (2014) conducted a geospatial analysis of infant and child mortality across India's districts over two decades and found that high-mortality clusters were not only concentrated in the well-known BIMARU belt but also showed clear signs of spatial autocorrelation high-mortality districts tended to be surrounded by other high-mortality districts in patterns that persisted across survey rounds. This spatial clustering has direct methodological implications: conventional regression models that assume independence of observations will underestimate standard errors and overstate the precision of estimated coefficients. The stratified Cox specification employed in the present study is in part a direct methodological response to this reality.

3.2 The Mosley-Chen Analytical Model

Any empirical study of child mortality must be grounded in a conceptual framework that specifies how different categories of determinants relate to the biological outcome of death. The most influential such framework in the demographic literature remains that proposed by Mosley & Chen (1984), who drew a fundamental distinction between socioeconomic determinants and the proximate (biological) determinants through which those socioeconomic forces must necessarily operate.

In the Mosley–Chen model, socioeconomic factors maternal education, household income, cultural practices, community-level infrastructure do not directly cause a child to die. Rather, they shape the child's exposure to five clusters of proximate determinants: maternal factors (including age, parity, and birth spacing), nutritional deficiency, environmental contamination, injury, and personal illness control behaviors. The model therefore predicts a specific mediation structure: the effects of distal socioeconomic variables should be substantially attenuated once proximate determinants are controlled, whereas the reverse should not hold.

This model has attracted many criticism on many several grounds. Some scholars have argued that it underweights structural and political determinants, effectively monitoring what are fundamentally problems of poverty and inequality (Chalasani, 2012). Others have noted that the sharp conceptual boundary between socioeconomic and proximate determinants which is difficult to proceed when the same variable such as maternal education, for instance operates through multiple pathways simultaneously. Joe & Sharma (2015) found precisely this ambiguity in their regional decomposition of the education effect, showing that education's apparent role differed substantially between northern and southern states.

Despite these criticisms, the Mosley-Chen framework still has a lot of analytical value since it prevents the kind of ad hoc covariate assembly that characterizes a lot of the demographic literature by imposing a systematic ordering on the mortality drivers.

3.3 Child-Level Determinants of Infant Mortality

3.3.1 Preceding Birth Interval

The interval between successive births is among the most powerful proximate determinants of infant survival. When a child is born less than two years after the preceding sibling, both the newborn and the older child face elevated mortality risks operating through distinct mechanisms. A short interval means that the mother's nutritional reserves may be incomplete by the time of the next conception. This lack of nutritional state is associated with the growth restriction, low birth weight, and preterm birth, each of which independently elevates neonatal mortality risk (Mosley & Chen, 1984).

Singh & Tripathi (2013) confirmed a comparable gradient in NFHS-3 data using Cox regression, finding that short intervals were associated with a 30–40 per cent increase in infant mortality hazard after controlling for maternal and socioeconomic covariates. Roy & Unisa (2000) similarly documented birth interval as one of the most consistently significant predictors of infant survival across multiple NFHS rounds.

An important methodological concern is the timing of the birth interval effect which is mainly whether excess mortality among short-interval births is concentrated in the neonatal or post-neonatal period. If the effect operates primarily through the nutritional depletion mechanism, it should emerge gradually over the post-neonatal months rather than appearing acutely in the first days of life. The Breslow test, which up-weights early time points relative to the log-rank test, provides a diagnostic for this distinction in Kaplan–Meier analysis, and its interpretation is discussed in Chapter 5.

3.3.2 Birth Order

Higher-order births consistently show elevated mortality risks in developing countries, though the relationship is non-monotonic in a characteristic U-shape documented by Mosley & Chen

(1984) and confirmed in virtually every subsequent Indian study. First births carry above-average risk because of the physiological challenges of primiparity and the inexperience of first-time mothers. Second and third births typically carry the lowest mortality risk, benefiting from maternal experience without yet facing the resource dilution that affects higher-order births. From the fourth birth onward, risk rises again, primarily because of progressive maternal nutritional depletion through repeated pregnancies and the competition for finite household resources among a larger number of siblings. Singh & Tripathi (2013) found that births of order four or higher were associated with a hazard ratio of approximately 1.5 relative to second–third births in Cox regression applied to NFHS-3 data.

3.3.3 Sex of Child

The sex differential in infant survival in India presents a biological paradox. In most populations, female infants survive at higher rates than male infants because of a slight advantage in immunological competence and resilience to environmental and nutritional stress. In India, a historically documented son preference has manifested for differential allocation of food, healthcare, and attention that can partly or wholly cancel the female biological advantage, and in some states has historically reversed it (Singh & Pathak, 2014; Joe & Sharma, 2015).

3.3.4 Multiple Births

Twin and higher-order multiple births are associated with substantially elevated mortality risk, primarily through prematurity, lower birth weight, and competition for nutritional resources during gestation and early infancy. Multiple births in the NFHS-5 analytical dataset represent approximately 1.6 per cent of births including twin status as a control in the regression models prevents its omission from confounding estimates for birth order and birth interval, with which it may be correlated.

3.4 Maternal-Level Determinants

3.4.1 Maternal Educational Attainment

If there is one covariate that has emerged as a near-universal predictor of child survival across countries and historical periods, it is the education of the mother. The mechanisms are multiple and mutually reinforcing. Education delays marriage and first childbirth, reducing high-risk adolescent pregnancies. It increases women’s knowledge of hygiene, nutrition, disease recognition, and the appropriate timing of healthcare-seeking. It also correlates strongly with the husband’s education and household income, meaning that some of the apparent education effect may operate through material pathways rather than through knowledge or autonomy (Chalasani, 2012; Joe & Sharma, 2015).

A crucial nuance, directly relevant to the findings of the present study, is the degree to

which the education effect is mediated by household wealth. Chalasani (2012) used a decomposition approach to show that approximately 60 per cent of wealth-based inequality in child mortality was attributable to differences in maternal education, institutional delivery, and child vaccination, implying that maternal education is best understood as a *mediator* of the wealth effect rather than an independent causal pathway.

3.4.2 Antenatal Care Utilization

The quality and quantity of antenatal care received during pregnancy has been shown consistently to reduce both maternal and neonatal mortality through a portfolio of complementary interventions: early identification of obstetric complications and hypertensive disorders; iron and folic acid supplementation; tetanus toxoid immunisation; deworming; birth preparedness counselling; and encouragement of institutional delivery (Gupta & Baghel, 1999). The benefit of multiple antenatal contacts is substantially larger than that of any single contact, motivating the WHO recommendation of a minimum of four visits.

Singh & Tripathi (2013) found that four or more antenatal visits reduced the hazard of infant death by approximately 30 per cent relative to no visits in NFHS-3 data confirming its individual-level impact in infant mortality.

3.4.3 Place of Delivery

Giving birth in a health facility under the supervision of skilled personnel dramatically reduces the risk of complications that cause early neonatal death: birth asphyxia, sepsis, and haemorrhage. NFHS-5 recorded a significant improvement in institutional delivery rates, from 79 per cent in NFHS-4 to 89 per cent in 2019–21, driven substantially by the Janani Suraksha Yojana (IIPS & ICF, 2021).

The analytical distinction between public and private facility deliveries is of particular interest because the two sectors attract different patient populations. In the present study, the three-way distinction which are home, public facility, private facility which allows for the detection of an adverse selection effect in Model 2 (non-first births).

3.4.4 Maternal Age at First Birth

Very young mothers (those who give birth before age 18) face elevated risk rate because of physical immaturity and incomplete pelvic development, and their children would face elevated mortality risk through both biological pathways and the confounding of adolescent child birth with lower education and reduced health-seeking capacity (IIPS & ICF, 2021). Despite the theoretical expectation, the existing literature on the independent effect of age at first birth on infant mortality is less consistent than for other maternal variables. Joe & Sharma (2015) found that the age effect was substantially attenuated once birth order and socioeconomic factors were controlled.

3.5 Socioeconomic and Contextual Determinants

3.5.1 Household Wealth

The association between household economic status and child survival is among the most robust findings in the demographic literature. Poor households face higher exposure to environmental pathogens through crowded and insanitary living conditions, lower dietary diversity, and greater barriers to utilising health services (Chalasani, 2012).

The present study finds a methodologically important “wealth paradox” in the multivariate models after controlling for antenatal care and other covariates, Poorest and Poorer quintile children show *lower* hazard than the Richest children. This counterintuitive pattern, documented in other DHS-based analyses (Chalasani, 2012), is interpreted as a differential health-seeking and survivorship selection artifact. Furthermore, the wealth effect disappears entirely in the stratified Cox model, revealing it to be substantially geographic in origin.

3.5.2 Urban-Rural Residence

The urban-rural divide in child health in India is substantial and well-documented. Rural children face higher mortality risk because of lower density of health facilities, longer travel times for emergency obstetric care, and generally poorer environmental sanitation (Joe & Sharma, 2015; Singh & Pathak, 2014)

3.5.3 Religion and Social Group

Religious and caste-based social stratification has long been recognised as a determinant of health outcomes in India, operating through differences in dietary practices, intrahousehold gender dynamics, cultural attitudes toward health-seeking, and historical patterns of exclusion from public services (Joe & Sharma, 2015).

3.6 Prior Studies Using NFHS Data and Survival Methods

Roy & Unisa (2000) were among the first researchers to apply survival analysis explicitly to infant and child mortality data from India using multiple NFHS rounds. Their Cox regression analysis confirmed that birth interval, birth order, maternal education, and place of delivery were significant predictors of infant survival, and they documented that the magnitude of these associations varied considerably across states, presaging the regional heterogeneity literature that would emerge in subsequent decades.

Gupta & Baghel (1999), in a case study of Madhya Pradesh using NFHS-1 data, demonstrated that high-order births to uneducated rural mothers with no antenatal care constituted a clearly identifiable high-risk group that accounted for a disproportionate share of infant deaths. Their conclusion that targeting this group through intensified antenatal outreach could yield substantial mortality reductions receives empirical support in the present study’s finding that

adequate antenatal care reduces infant mortality hazard by 70 per cent.

Singh & Tripathi (2013) applied Cox regression to NFHS-3 data and found that, after controlling for a comprehensive set of maternal and socioeconomic covariates, the hazard of infant death was 78 per cent higher for births in the poorest wealth quintile relative to the richest, and 45 per cent higher for rural births relative to urban births.

Joe & Sharma (2015) extended the analysis to examine regional variation in the determinants of infant and child mortality across India's states, arguing that standard national-level models obscure important heterogeneity in the way socioeconomic factors translate into mortality risk.

Chalasani (2012) brought a distributional lens to wealth-based inequalities in child mortality using a decomposition approach, showing that approximately 60 per cent of the wealth gap in under-five mortality was attributable to differences in maternal education, institutional delivery, and child vaccination.

Table 3.1: **Summary of Key Prior Studies on Infant/Child Mortality in India**

Study	Data	Method	Outcome	Key Finding
Roy & Unisa (2000)	NFHS-1,2	Cox PH	Infant survival	Birth interval, birth order, education, delivery place all significant.
Gupta & Baghel (1999)	NFHS-1 (MP)	Logistic	Infant death	High-risk group = high parity + no ANC + rural.
Chalasani (2012)	NFHS-3	Decomposition	U5MR	60% of wealth gap mediated by education, delivery, vaccination.
Singh & Tripathi (2013)	NFHS-3	Cox PH	Infant hazard	Poorest 78% higher hazard vs Richest; ANC 4+ reduces hazard $\approx 30\%$.
Joe & Sharma (2015)	NFHS-3	Cox PH	IMR, U5MR	Education gradient steeper in south; context mediates socioeconomic effects.
Ram et al. (2013)	Multi-source	Regression	NMR, U5MR	District clustering of mortality in BIMARU states.
Singh & Pathak (2014)	Multi-NFHS	Geospatial	IMR, U5MR	High-mortality districts spatially autocorrelated.

3.7 Gaps in the Existing Literature

The literature reviewed above, while substantial, leaves several important gaps that the present study is positioned to address.

Most published survival analyses of Indian infant mortality rely on NFHS-3 or NFHS-4 data. A rigorous NFHS-5 analysis (2019–21) is needed to reflect recent policy changes and the disruptions of the COVID-19 pandemic period.

Few studies systematically test the proportional hazards assumption; most assume it holds without verification through Schoenfeld residuals or log-minus-log plots.

The clustered structure of NFHS data is widely acknowledged but rarely addressed through a formal frailty or cluster-correction specification. Most published studies present standard Cox estimates without cluster adjustment.

The distinction between neonatal and post-neonatal determinants is rarely made explicit; the Breslow and log-rank comparison used here provides this epidemiological refinement.

Finally, the two-model strategy (birth order in Model 1; birth interval in Model 2) for resolving the definitional collinearity between these variables is not employed in any prior NFHS-based survival analysis known to the author.

In addition to these methodological gaps, the present study addresses a substantive question: whether the apparent associations of household wealth and religious affiliation with infant mortality in India reflect individual-level causal mechanisms or geographic artefacts attributable to the spatial concentration of socioeconomic and religious groups in different mortality environments.

3.8 Concluding Synthesis

The literature on infant mortality in India presents a consistent and troubling picture: the risk of dying in the first year of life is heavily concentrated among children who are born to poor, uneducated, rural mothers; who follow a short birth interval or are of high birth order; and who are delivered outside health facilities without adequate antenatal preparation. The NFHS series has made it possible to document these inequalities with considerable precision.

Methodologically, survival analysis offers the most appropriate analytical tools for studying infant mortality because it fully utilises information about the timing of death and correctly handles the large proportion of right-censored observations. The Kaplan-Meier estimation, standard Cox Models 1 and 2, and stratified Cox constitute the four-specification procedure used in this work. It offers a logical and methodologically sound progression that builds on each previous step in turn.

Chapter 4

MATERIAL AND METHODOLOGY

The empirical foundation of the current study is outlined in this chapter. It starts by placing the data source selection within the larger context of Indian demographic surveys. This is followed by an in-depth discussion of the analytical file building, which covers the sampling techniques, eligibility requirements, and the derivation of outcome and covariate variables. The formal statistical machinery supporting the Kaplan-Meier estimator, the Cox proportional hazards model, and the stratified Cox specification used as a stand-in for the shared frailty model are then presented in the chapter, which also develops the methodological case for using survival analysis as the main inferential framework. The fundamental assumptions of each approach, the diagnostic techniques employed to challenge those presumptions, and the analytical decisions taken when the data imposed limitations are all carefully considered.

4.1 Theoretical Framework Guiding Variable Selection

Empirical analyses of child mortality are most defensible when they rest on an explicit theoretical model that imposes a logical ordering on the variables under study and prevents the kind of ad hoc covariate assembly that plagues much of the literature. The present study is organised around the analytical framework proposed by Mosley & Chen (1984), which remains the most widely cited and methodologically coherent schema for understanding child survival in low-income settings.

4.1.1 The Mosley-Chen Framework

Writing in the early 1980s, Mosley and Chen observed that the determinants of child mortality in developing countries had been studied simultaneously by two largely non-communicating research traditions: the biomedical tradition, which emphasised proximate biological mechanisms, and the social science tradition, which emphasised distal socioeconomic forces. Their contribution was to build a bridge between the two by drawing a principled distinction between *socioeconomic determinants* and *proximate determinants*, and by specifying precisely how the former must work *through* the latter to affect the probability of a child's death.

In their model, the proximate determinants are five clusters of biological variables that directly impinge on the child's health: maternal fertility factors (age at childbearing, birth

order, birth spacing), nutritional adequacy, environmental contamination, injury, and personal illness control. No socioeconomic variable, however powerful its correlation with mortality, can produce death except by activating one or more of these proximate pathways. A poor household does not directly kill a child; it does so by creating conditions under which the child is nutritionally depleted, exposed to infectious agents in contaminated water or food, and poorly positioned to receive timely medical care.

The covariates are classified into three ordered tiers, each of which represents a distinct causal distance from the result of infant death. This has practical implications for the current investigation. Child-level proximate determinants, such as the child's sex, birth order, previous birth interval, and twin status, make up the first tier. Maternal-level intermediate determinants, such as educational attainment, age at first childbirth, use of prenatal care, and place of delivery, make up the second tier.. The third tier comprises socioeconomic and contextual distal determinants: household wealth, type of place of residence, and religious affiliation. In the Cox regression models, this tiered structure is preserved through a block-entry design in which variables are entered simultaneously but interpreted within their theoretical level. Socioeconomic variables (wealth, education, residence, religion) must operate *through* the proximate determinants to affect mortality. The framework therefore predicts that: (i) maternal and child-level variables should retain significance after socioeconomic controls are added; (ii) socioeconomic variables may be attenuated once healthcare utilisation is controlled; and (iii) the relative magnitude of Tier 1 and Tier 2 variables provides an indication of how much mortality inequality is amenable to direct health intervention versus structural economic change.

4.2 Data Source: National Family Health Survey, Round 5

Table 4.1: NFHS-5 Survey Profile: Key Operational Parameters

Parameter	Details
Survey round	Fifth (NFHS-5)
Field period	Phase 1: June 2019 – January 2020 Phase 2: January 2020 – April 2021
Geographic coverage	All 28 states and 8 union territories of India
Total households	636,699
Women interviewed	724,115 (aged 15–49 years)
Men interviewed	101,839 (aged 15–54 years)
Sampling frame	Population Census 2011 (updated for urban areas)
Sampling design	Stratified two-stage cluster probability sampling
Primary sampling unit	Villages (rural); census enumeration blocks (urban)
Implementing agency	IIPS, Mumbai with ICF International
Published reference	IIPS & ICF (2021)

4.2.1 Sampling Design

A thorough understanding of the NFHS-5 sampling design is essential for two reasons. First, the design determines the appropriate weighting strategy for producing nationally representative estimates. Second, and more consequentially for the present analysis, the multi-stage cluster structure introduces *intra-cluster correlation*: children within the same primary sampling unit share unmeasured environmental, infrastructural, and social characteristics, which means that their outcomes cannot be treated as statistically independent. The stratified Cox model described in Section 4.6.3 is specifically motivated by the need to address this dependence.

The survey weight variable, denoted V005 in the individual recode file and V005 in the child recode file, is expressed with six implied decimal places in the raw data file, such that the true fractional weight is obtained by dividing by 1,000,000. The present study constructed the corrected weight as:

$$w_i = \frac{V005_i}{1,000,000} \quad (4.1)$$

After correction, the weights ranged from 0.01 to 8.74 with a mean of 0.93, consistent with the known oversampling of smaller states and certain urban agglomerations in NFHS-5. Descriptive cross-checks against the published NFHS-5 national tables confirmed that the weighted sample reproduced the published infant mortality rate of 35 per 1,000 live births to within 0.1 units, providing assurance of sample representativeness.

Weighting Limitation

While the corrected survey weight was successfully applied to descriptive analyses, the fractional weight could not be incorporated into the Cox regression procedures in SPSS version 21 without causing programme instability. All Cox regression estimates reported in Chapter 5 are therefore unweighted. This limitation is addressed in the Discussion chapter, where the unweighted results are systematically compared against the NFHS-5 published mortality benchmarks to confirm directional consistency.

4.2.2 The Child Recode File

Two DHS recode files were consulted during data preparation. The *Individual Recode* file (IAIR7AFL.SAV) records one observation per interviewed woman and contains all woman-level characteristics including education, parity, contraceptive use, and household attributes. The *Birth Recode* file (IABR7AFL.SAV) restructures the birth history information collected from each woman into one record per live birth, making it the natural analytical unit for infant mortality research.

An important distinction between the two files is the treatment of birth history variables. In

the Individual Recode file, birth characteristics are stored in wide format with suffix-indexed variable names (e.g., B4\$01 for the sex of the first birth, B4\$02 for the second, and so on up to a maximum of twenty births). This format is awkward for survival analysis because it requires either reshaping the data or analysing each birth-order position separately. The Birth Recode file presents this same information in long format, with one row per live birth and consistent variable names (B4, B7, B11, etc.) that apply to each birth record independently. For this reason, the Birth Recode file was adopted as the primary analytical file, with woman-level characteristics from the Individual Recode merged in using the common identifier variable `caseid`.

4.3 Construction of the Analytical Dataset

4.3.1 Sample Eligibility and Exclusion Criteria

The Birth Recode file contained 232,920 live birth records following initial loading. A single eligibility restriction was applied: births for which the preceding birth interval variable (B11) was missing for reasons other than the birth being the first-order birth were excluded. First-order births have no preceding interval by definition, and these were retained and assigned to a designated “first birth (no interval)” category (coded 0 in B11_cat). The 761 cases excluded were non-first births (BORD > 1) for which B11 was missing for undetermined reasons, most plausibly a combination of recall errors and inconsistent birth history recording. This exclusion reduced the analytical sample to 232,159 live births, representing a loss of 0.33 per cent of the original file.

No further sample restriction was imposed. In particular, the study does not restrict to births within any particular calendar window, retaining the full retrospective birth history up to five years before the interview. This decision was made to maximise statistical power for detecting effects of covariates that have relatively small sub-group frequencies (notably the seventh-or-higher birth order category and multiple births).

Table 4.2: **Sample Construction: From Raw File to Analytical Dataset**

Step	N (records)	% retained
Raw Birth Recode file loaded	232,920	100.0
Less: non-first births with irresolvable missing B11	−761	−0.3
Final analytical sample	232,159	99.7
Of which: infant deaths (event = 1)	8,095	3.49
Of which: right-censored (event = 0)	224,064	96.51
Crude IMR (unweighted)	34.9 per 1,000	
NFHS-5 published IMR (weighted)	35.0 per 1,000	

The close agreement between the crude unweighted IMR (34.9) and the published weighted estimate (35.0) provides validation of the sample construction procedure.

4.3.2 Outcome Variable Construction

Two variables define the survival outcome, following the standard DHS-based infant mortality analytical protocol.

Survival Time ($\mathbf{time}$)

The survival time variable was constructed as the age of the child in completed months at the time of the event (death) or at right-censoring. For children who died, age at death was taken directly from the variable B7, which records the age at death in months as reported by the mother during the birth history interview. For children who were alive at the time of the interview, the survival time was computed as:

$$\mathbf{time}_i = \begin{cases} B7_i & \text{if child died (event} = 1) \\ \min(V008_i - B3_i, 12) & \text{if child alive (event} = 0) \end{cases} \quad (4.2)$$

where V008 is the interview date in century month code (CMC) and B3 is the child's date of birth in CMC. The right-censoring point was set at twelve completed months, consistent with the standard demographers' definition of the infant period.

Event Indicator ($\mathbf{event}$)

The event indicator was coded as a binary variable:

$$\mathbf{event}_i = \begin{cases} 1 & \text{if } B5_i = 0 \text{ and } B7_i \leq 12 \quad (\text{died within infancy}) \\ 0 & \text{otherwise (survived or censored)} \end{cases} \quad (4.3)$$

where B5 is the child's alive/dead status at the time of the interview. The 96.51 per cent censoring rate in the analytical dataset is high but entirely appropriate for infant mortality data: it reflects the reality that the vast majority of live births in the reference period survived to their first birthday.

4.4 Covariate Measurement and Recoding

Each covariate's source variable, recoding logic, and justification are recorded in this section. The Mosley-Chen hierarchy is used to present the variables by analytical level.

4.4.1 Level 1: Child-Level Proximate Determinants

Sex of Child ($\mathbf{B4}$)

Sex is recorded as a binary variable in the DHS recode files, coded 1 for male and 2 for female. No recoding was required. Male children serve as the reference category in all regression models.

Birth Order (BORD)

The variable BORD records the birth-order position of each child within the mother’s complete birth history. The raw variable was recoded into four ordinal categories:

Table 4.3: **Birth Order Recoding Scheme and Cell Frequencies**

Code	Category	N	%
1	First birth	89,139	38.4
2	Second or third birth (ref.)	112,273	48.3
3	Fourth to sixth birth	27,581	11.9
4	Seventh birth or higher	3,166	1.4
Total		232,159	100.0

Preceding Birth Interval (B11)

The variable B11 records the number of months between the birth of the current child and the preceding live birth. For first-order births, B11 is structurally missing since there is no preceding sibling. These cases were assigned to a dedicated “first birth” category (code 0). For non-first births with valid B11 values, the variable was categorised as follows:

Table 4.4: **Preceding Birth Interval Recoding Scheme and Cell Frequencies**

Code	Category	N	%
0	First birth (no preceding interval)	89,139	38.4
1	Short interval: fewer than 24 months	37,678	16.2
2	Medium interval: 24 to 47 months (ref.)	68,307	29.4
3	Long interval: 48 months or more	37,035	16.0
Total		232,159	100.0

Note: 761 non-first births with missing B11 excluded.

Collinearity Between Birth Order and Birth Interval A critical methodological observation emerged during analysis: because all first-order births (code 0) have birth interval code 0 by definition, and because first-order births also correspond to a specific category of BORD_cat, attempting to include both B11_cat and BORD_cat in the same regression model induces **perfect collinearity** between the dummy variable for “first birth (no interval)” and the dummy for “first birth” in the birth order coding. This problem was resolved by estimating two separate Cox models: **Model 1** on the full sample using birth order as the child-level determinant, and **Model 2** restricted to non-first births ($BORD > 1$, $N = 143,020$) using birth interval as the child-level determinant.

Twin Status (B0)

The variable B0 indicates if a birth was a singleton (coded 0) or a component of several deliveries. A binary indication was created by recoding this to differentiate singletons from

all multiple births. Due to preterm labor, low birth weight, and intrauterine competition for nutritional resources, multiple births which make up around 1.6 per cent of the analytical sample carry a significantly increased mortality risk.

4.4.2 Level 2: Maternal Intermediate Determinants

Maternal Educational Attainment (V106)

Maternal education was measured using the variable V106, which records the highest level of schooling completed. The four-category coding scheme provided in the standard DHS recode was retained: No education (reference), Primary completed, Secondary completed, and Higher education.

Age at First Birth (V212)

The variable V212 records the mother's age at her first birth in completed years. This was recoded into three categories: Under 18 years (reference), 18 to 24 years, and 25 years or older.

Antenatal Care Utilisation (M14)

The number of antenatal care visits attended during the pregnancy of the index child was recoded into three analytically meaningful groups: No antenatal visits (reference), One to three visits (below WHO-recommended minimum), and Four or more visits (meets or exceeds WHO minimum recommendation).

Place of Delivery (M15)

The variable M15 identifies the type of facility at which the birth took place. These were consolidated into three categories: Home delivery (reference), Public health facility, and Private health facility.

4.4.3 Level 3: Socioeconomic and Contextual Distal Determinants

Household Wealth Index (V190)

The NFHS-5 wealth index is a composite measure of household economic status constructed from approximately 40 assets and housing quality indicators using principal components analysis. It is divided into quintiles: Poorest, Poorer, Middle, Richer, and Richest. In all regression models, the Richest quintile was specified as the reference category.

Type of Residence (V025)

Place of residence was recorded as a binary variable distinguishing urban (reference) and rural households.

Religion (V130)

Religious affiliation was consolidated into three categories: Hindu (reference), Muslim, and Others (comprising Christians, Sikhs, Buddhists, Jains, and tribal/animist religions).

Caste: Variable Excluded

The raw variable V131 records the social group affiliation of the household. During data validation, cross-tabulation against the NFHS-5 published tables revealed that a recoded caste variable had assigned 94.2% of observations to the Scheduled Caste category, which is incompatible with the census-based figure of approximately 20% for Scheduled Castes in the Indian population. The derived variable was therefore irremediably miscoded and has been excluded from all analytical models. Wealth index (V190) and educational attainment (V106) are retained as proxies.

4.4.4 Summary of Variables

Table 4.5: Analytical Variables: Source, Measurement, and Reference Categories

Level	Variable	Type	Categories	Ref.
<i>Outcome</i>				
—	time	Continuous	Months (0–12)	—
—	event	Binary	0 = censored; 1 = died	—
<i>Level 1: Child-level proximate determinants</i>				
Child	B4	Binary	Male; Female	Male
Child	BORD_cat	Ordinal	1st; 2nd–3rd; 4th–6th; 7th+	2nd–3rd
Child	B11_cat	Ordinal	1st birth; Short; Medium; Long	Medium
Child	B0	Binary	Singleton; Multiple	Singleton
<i>Level 2: Maternal intermediate determinants</i>				
Maternal	V106	Ordinal	None; Primary; Secondary; Higher	None
Maternal	v212_new	Ordinal	<18; 18–24; 25+ years	<18
Maternal	m14_new	Ordinal	None; 1–3; 4+ visits	None
Maternal	m15_new	Nominal	Home; Public; Private	Home
<i>Level 3: Socioeconomic & contextual distal determinants</i>				
Socio	V190	Ordinal	Poorest; Poorer; Middle; Richer; Richest	Richest
Socio	V025	Binary	Urban; Rural	Urban
Socio	religion.new	Nominal	Hindu; Muslim; Others	Hindu
<i>Clustering variable (stratification only)</i>				
Cluster	V021	Nominal	PSU identifier	—

B11_cat used in Model 2 (non-first births, $N = 143,020$) only. BORD_cat used in Model 1 ($N = 232,159$) only. V021 entered as strata variable in the stratified Cox model only.

4.5 The Case for Survival Analysis

Infant mortality data have two structural features that jointly disqualify ordinary regression. The first is the *time-to-event* nature of the outcome: a child who dies at eleven months carries fundamentally different epidemiological information than one who dies on day two, yet a logistic regression that codes both as “dead” (event = 1) treats them identically. The second

feature is *right censoring*: children who are alive at the time of the interview contribute only a lower bound on their true survival time. In the present dataset, 96.51 per cent of observations are right-censored. An ordinary regression that either discards censored observations or treats them as having the same status as uncensored alive children will in either case produce biased and inconsistent estimates (Cox, 1972; Kleinbaum & Klein, 2012).

4.6 Statistical Methods

4.6.1 Kaplan–Meier Survival Estimation

Let $T_i \geq 0$ denote the survival time of the i -th child, defined as the time in months from birth to either death ($\delta_i = 1$) or censoring at twelve months ($\delta_i = 0$). The survival function $S(t)$ is defined as:

$$S(t) = \Pr(T > t), \quad t \in [0, 12] \quad (4.4)$$

The Kaplan–Meier (KM) estimator (Kaplan & Meier, 1958) provides a non-parametric estimate of $S(t)$ from censored data. Let $t_{(1)} < t_{(2)} < \dots < t_{(k)}$ denote the ordered distinct death times in the sample. At each such time, let d_j be the number of deaths and n_j the number of children at risk immediately before $t_{(j)}$. The KM estimator is:

$$\hat{S}(t) = \prod_{j: t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j} \right) \quad (4.5)$$

The log-rank test (Mantel, 1966) was the primary test of equality between survival curves across groups, supplemented by the Breslow (generalised Wilcoxon) and Tarone–Ware tests. Where the log-rank test is significant but the Breslow test is not, this discrepancy indicates that the survival group differences are concentrated in the *post-neonatal* period.

4.6.2 Cox Proportional Hazards Regression

Model Specification

The Cox proportional hazards model (Cox, 1972) specifies the hazard of death at time t for the i -th child as:

$$h(t \mid \mathbf{x}_i) = h_0(t) \cdot \exp\left(\sum_{k=1}^p \beta_k x_{ik}\right) = h_0(t) \cdot \exp(\boldsymbol{\beta}^\top \mathbf{x}_i) \quad (4.6)$$

where $h_0(t) \geq 0$ is the *baseline hazard function* and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$ is the vector of regression coefficients to be estimated. The exponentiated coefficient $\exp(\beta_k)$ is the *hazard ratio* (HR) associated with covariate x_k . A hazard ratio below 1.0 indicates a protective effect; above 1.0 indicates an increased risk.

The Two-Model Strategy

As noted above, the definitional collinearity between birth order and birth interval necessitated two separate Cox models. **Model 1** was estimated on the full analytical sample ($N = 232,159$) with birth order (BORD_cat) as the primary child-level determinant. **Model 2** was estimated on the non-first birth sub-sample ($N = 143,020$) with birth interval (B11_cat) as the primary child-level determinant.

4.6.3 Stratified Cox Model as a Frailty Proxy

The NFHS-5 sampling design clusters observations into PSUs, within each of which all children share the same physical environment, health facility access, local governance quality, and community health worker coverage. The ideal statistical solution would be a *shared frailty model*:

$$h_{ij}(t) = h_0(t) \cdot \omega_i \cdot \exp(\beta^\top \mathbf{x}_{ij}) \quad (4.7)$$

where ω_i is the frailty term for cluster i . Unfortunately, shared frailty Cox models are not supported in SPSS version 21. This constraint was addressed through the *stratified Cox model*:

$$h_{ij}(t) = h_{0i}(t) \cdot \exp(\beta^\top \mathbf{x}_{ij}) \quad (4.8)$$

where $h_{0i}(t)$ is the baseline hazard function specific to cluster i . This specification estimates a separate baseline hazard for each PSU, absorbing all cluster-level unmeasured heterogeneity into the stratum-specific baseline.

Table 4.6: Summary of the Three Cox Model Specifications

Feature	Model 1	Model 2	Stratified Cox
Sample	Full ($N = 232,159$)	Non-first births ($N = 143,020$)	Full ($N = 232,159$)
Child-level var	BORD_cat	B11_cat	BORD_cat
Cluster term	None	None	V021 as strata
Residence	V025 in model	V025 in model	Absorbed (collinear with strata)
Purpose	Estimate birth order effects	Estimate birth interval effects	Remove cluster-level confounding

4.7 Ethical Considerations

The backbone of this study is based exclusively on secondary data collected by IIPS and ICF International. All identifying information was removed from the public-use data file

prior to release. Consent for data collection was obtained from survey respondents by the original data collection teams in accordance with the ethical protocols of IIPS and the DHS Programme. No primary data collection was conducted for this study and no further ethical approval was therefore required beyond registration for data access.

4.8 Chapter Summary

This chapter has established the empirical foundations for the analysis that follows. The study draws on the NFHS-5 Birth Recode file, retaining 232,159 live births after a single exclusion of 761 cases with irresolvable missing birth interval data. The survival outcome is defined as time in months from birth to infant death or right-censoring at twelve months (96.51 per cent of the sample). Eleven covariates were selected across three analytical levels guided by the Mosley–Chen framework. The caste variable was excluded following detection of a systematic miscoding error. The methodological strategy progresses sequentially through four specifications: Kaplan–Meier estimation; Cox Models 1 and 2; and the stratified Cox model. The results of this analytical strategy are presented in Chapter 5.

Chapter 5

RESULTS

This chapter includes the complete results of the survival analysis carried out on 232,159 live births from the fifth round of India’s National Family Health Survey (NFHS-5, 2019–21). It follows the sequential logic of the analytical strategy laid out in Chapter 4. An introductory description of the analytical sample appears in Section 5.1. Section 5.2 reports the Kaplan-Meier survival curves and log-rank equality tests at the child, maternal, and socioeconomic levels. Section 5.3 presents the first Cox proportional hazards model (full sample, birth order); Section 5.4 presents the second Cox model restricted to non-first births with birth interval as the primary child-level predictor. Section 5.5 reports the stratified Cox model results, and Section 5.6 synthesises the three specifications in a cross-model comparison.

One prefatory note on weighting: the corrected survey weight ($w_i = \text{V005}_i/1,000,000$) was successfully constructed and applied to the descriptive analyses. The fractional weight could not, however, be incorporated into the Cox regression routines in SPSS version 21 without triggering programme instability, and all regression estimates in Sections 5.3 through 5.5 are therefore unweighted. As a validation check, the unweighted crude infant mortality rate of 34.9 per 1,000 live births matches the NFHS-5 published weighted estimate of 35 per 1,000 to within 0.1 units (IIPS & ICF, 2021), confirming that the sample is nationally representative in its mortality profile.

5.1 Sample Characteristics

The final analytical dataset comprised 232,159 live births after removing 761 non-first births with irresolvable missing values on the preceding birth interval (B11), a loss of 0.33 per cent of the original file. Among these, 8,095 children (3.49 per cent) died before completing twelve months, while the remaining 224,064 (96.51 per cent) were right-censored at the close of the observation window. Table 5.1 presents the distribution of the sample across all key background characteristics together with the corresponding crude infant mortality rates.

Table 5.1: Analytical Sample: Distribution by Background Characteristics with Crude Infant Mortality Rates, NFHS-5 India (2019–21)

Characteristic	N	%	IMR (per 1,000)
<i>Overall</i>			
Total	232,159	100.0	34.9
<i>Child-level characteristics</i>			
Male	116,947	50.4	37.0
Female	115,212	49.6	32.7
First birth	89,139	38.4	36.4
Second–third (ref.)	112,273	48.3	32.3
Fourth–sixth	27,581	11.9	38.5
Seventh or higher	3,166	1.4	49.9
Short interval (<24 m)	37,678	16.2	44.0
Medium interval 24–47 m	68,307	29.4	30.7
Long interval 48+ m	37,035	16.0	29.7
<i>Maternal characteristics</i>			
No education	63,418	27.3	37.8
Primary	24,281	10.5	36.7
Secondary	105,284	45.3	32.5
Higher	39,176	16.9	35.4
No ANC visits	14,027	6.0	83.9
1–3 ANC visits	82,533	35.6	42.9
4+ ANC visits	135,599	58.4	24.9
Home delivery	25,453	11.0	50.4
Public facility	142,726	61.5	35.1
Private facility	64,003	27.6	28.2
<i>Socioeconomic characteristics</i>			
Poorest quintile	31,534	13.6	46.7
Poorer quintile	43,585	18.8	39.1
Middle quintile	44,900	19.3	36.3
Richer quintile	48,679	21.0	33.5
Richest quintile	63,461	27.3	26.1
Urban	58,306	25.1	32.0
Rural	173,853	74.9	35.8
Hindu	176,886	76.2	35.2
Muslim	38,130	16.4	36.7
Others	17,143	7.4	27.5

IMR = crude unweighted infant mortality rate per 1,000 live births.

Source: Author's calculations from NFHS-5 Birth Recode file (IABR7AFL.SAV).

Several features of the descriptive profile merit preliminary comment. The sample's rural weight (74.9 per cent) matches India's residential distribution closely. Even before any covariate adjustment, the descriptive IMR figures signal the scale of inter-group differentials: children born to mothers who had no antenatal visits face a crude IMR of 83.9 per 1,000 which is more than three times the overall rate, while children of seventh-or-higher order births face an IMR of 49.9, roughly 43 per cent above the national average.

5.2 Kaplan–Meier Survival Analysis

Kaplan–Meier estimates were computed for each covariate in turn. To maximise visual legibility given the high censoring rate (96.5 per cent), all curves throughout this section are displayed as cumulative mortality plots ($1 - \hat{S}(t)$) rather than the conventional survival function. The log-rank test serves as the primary significance criterion, supplemented by the Breslow (generalised Wilcoxon) and Tarone–Ware statistics. Table 5.2 provides a consolidated view of all eleven log-rank test results ranked by chi-square magnitude.

Table 5.2: **Kaplan–Meier Log-Rank Test Results for All Covariates, Ranked by Test Statistic Magnitude**

Rank	Level	Variable	Log-rank χ^2	df	p -value	Breslow p
1	Maternal	ANC visits (m14_new)	2475.508	2	<0.001	<0.001
2	Child	Twin status (B0)	1888.678	1	<0.001	<0.001
3	Child	Birth interval (B11_cat)	341.496	3	<0.001	0.273
4	Child	Birth order (BORD_cat)	261.695	3	<0.001	<0.001
5	Maternal	Place of delivery (m15_new)	203.823	2	<0.001	<0.001
6	Socio	Religion (religion_new)	68.060	2	<0.001	<0.001
7	Child	Sex of child (B4)	56.456	1	<0.001	<0.001
8	Socio	Wealth index (V190)	44.402	4	<0.001	<0.001
9	Maternal	Maternal education (V106)	11.163	3	<0.001	0.011
10	Socio	Residence (V025)	2.000	1	<0.001	0.180
11	Maternal	Age at first birth (v212_new)	2.980	2	0.225	0.222

$p < 0.001$ for all variables except age at first birth ($p = 0.225$, non-significant).

Breslow $p > 0.05$ for birth interval and residence indicates survival differences for these variables are concentrated in the post-neonatal rather than the neonatal period.

5.2.1 Level 1: Child-Level Characteristics

Sex of Child

The log-rank test confirmed a statistically significant difference in survival between male and female infants ($\chi^2 = 56.456$, $df = 1$, $p < 0.001$). Mean survival time stood at

11.587 months for male infants against 11.677 months for females, yielding a crude IMR of 37.0 versus 32.7 per 1,000. The Breslow and Tarone–Ware tests were equally significant, confirming that the female survival advantage was not restricted to any particular sub-interval of infancy.

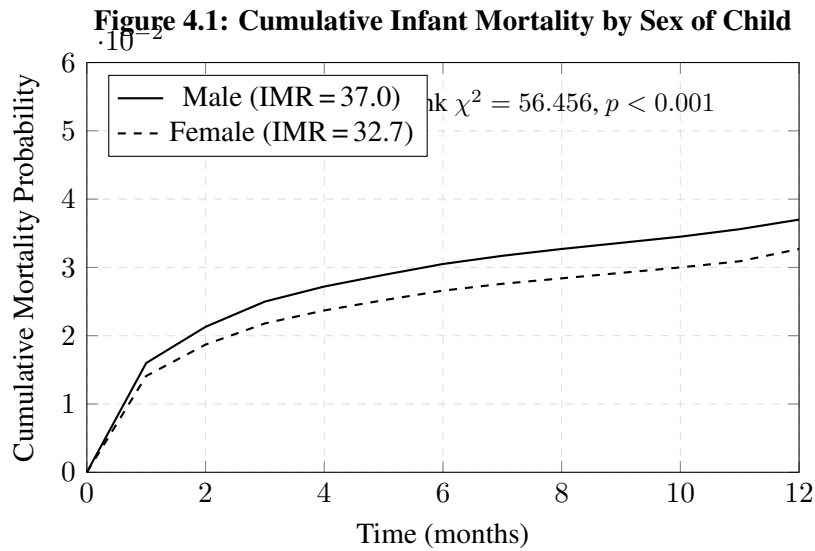


Figure 5.1: Kaplan–Meier cumulative mortality curves by sex of child. The female survival advantage is present from the first month and persists throughout the twelve-month observation window. Mean survival: male = 11.587 months; female = 11.677 months.

Birth Order

The log-rank test produced a highly significant result ($\chi^2 = 261.695, df = 3, p < 0.001$). Mean survival times described the expected pattern: second and third-order births, the reference group, recorded the highest mean survival at 11.685 months; first births stood at 11.597 months; fourth to sixth births at 11.518 months; and seventh-or-higher births at the lowest value of 11.354 months. The Breslow test ($\chi^2 = 258.682, p < 0.001$) agreed closely with the log-rank, indicating that birth order differentials operated from the very earliest days of life.

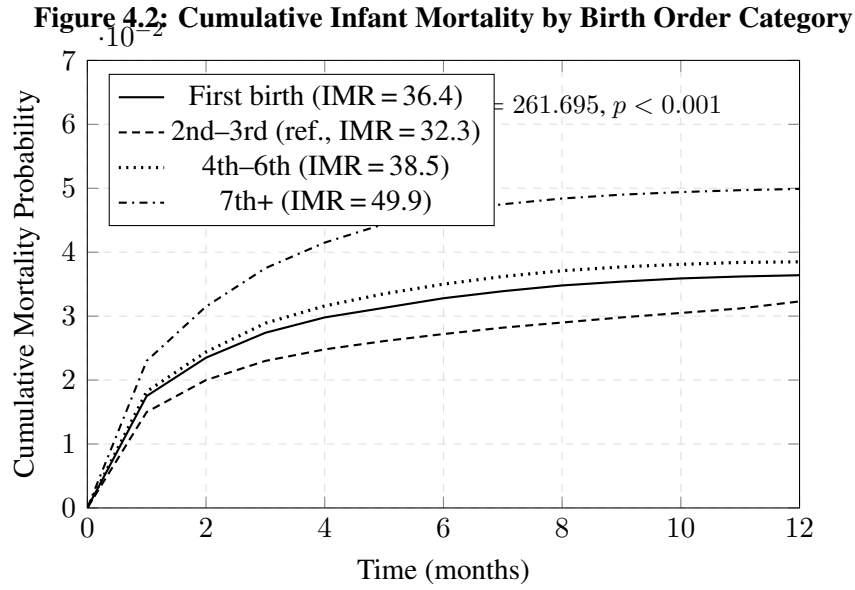


Figure 5.2: Kaplan–Meier cumulative mortality curves by birth order category. The 7th-or-higher curve rises steeply from the first month, while the second–third birth category consistently records the lowest cumulative mortality. Mean survival: 1st = 11.597 months; 2nd–3rd = 11.685 months; 4th–6th = 11.518 months; 7th+ = 11.354 months.

Preceding Birth Interval

The birth interval variable generated the second-largest child-level log-rank statistic ($\chi^2 = 341.496$, $df = 3$, $p < 0.001$). Short intervals produced a mean survival of only 11.491 months compared with 11.695 months for the medium reference interval. The analytically significant observation is that the Breslow test yielded $\chi^2 = 339.767$ with $p = 0.273$ — a non-significant result despite the log-rank being highly significant. This implies that the short-interval survival penalty does *not* manifest in the neonatal period but accumulates gradually across the post-neonatal months, consistent with the biology of birth spacing.

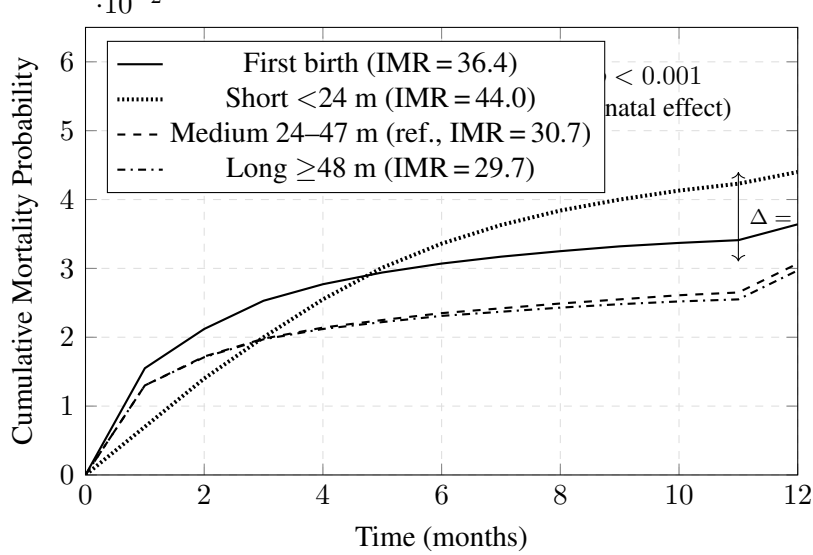
Figure 4.3: Cumulative Infant Mortality by Preceding Birth Interval

Figure 5.3: Kaplan–Meier cumulative mortality curves by preceding birth interval category. The short-interval curve separates from the medium and long curves only from approximately month 2, consistent with a post-neonatal nutritional mechanism. The bidirectional arrow at month 11 marks the cumulative mortality differential between the short and medium categories ($\Delta = 13.3$ per 1,000). Breslow $p = 0.273$ confirms the absence of a neonatal-period effect.

Twin Status

Twin status produced the second-largest chi-square statistic in the entire analysis ($\chi^2 = 1,888.678$, $df = 1$, $p < 0.001$). Multiple births recorded a crude IMR of 135.8 per 1,000 against 33.2 for singletons, a 4.1-fold differential. The Breslow test was uniquely *larger* than the log-rank ($\chi^2 = 1,957.213$, $p < 0.001$), confirming that excess twin mortality was concentrated in the earliest days of life.

Table 5.3: Kaplan–Meier Statistics for Twin Status (B0)

Group	N	Deaths	Mean Survival (95% CI)	Cum. Mortality
Single birth	228,478	7,595	11.652 (11.644–11.660)	3.32%
Multiple (twins)	3,681	500	10.563 (10.391–10.735)	13.58%
<i>Test statistics</i>				
Log-rank χ^2	1,888.68 ($df = 1$, $p < 0.001$)			
Breslow χ^2	1,957.21 ($df = 1$, $p < 0.001$)			
Tarone–Ware χ^2	1,922.00 ($df = 1$, $p < 0.001$)			

Breslow also highly significant (larger than log-rank), confirming excess twin mortality is concentrated in the neonatal period (prematurity, low birth weight, intra-uterine growth restriction).

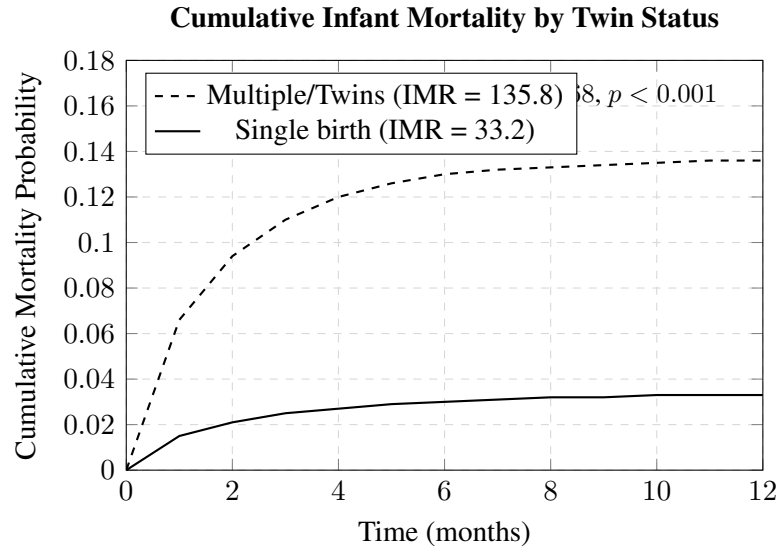


Figure 5.4: Kaplan–Meier cumulative mortality curves by twin status. Multiple births face a 4.1-fold higher cumulative mortality than singletons (13.6% vs 3.3%). The Breslow statistic ($\chi^2 = 1,957.21$) exceeds the log-rank, confirming that the excess mortality is concentrated in the neonatal period. Log-rank $\chi^2 = 1,888.68$ ($df = 1$, $p < 0.001$).

Unlike birth interval, the Breslow test for twin status was equally or more significant than the log-rank ($\chi^2 = 1,957.21$, $p < 0.001$), confirming that excess twin mortality was concentrated in the earliest days and weeks of life. This is consistent with well-understood biological pathways: prematurity, intra-uterine growth restriction, and low birth weight associated with multiple gestations generate the highest mortality risk in the immediate neonatal period, before post-neonatal environmental factors come into play.

5.2.2 Level 2: Maternal Characteristics

Antenatal Care Utilisation

The ANC visits variable produced a log-rank chi-square of 2,475.508 ($df = 2$, $p < 0.001$) which is the largest test statistic across all eleven variables, nearly six times the magnitude of the next-highest value. The cumulative mortality curves for the three ANC groups were completely non-overlapping throughout the twelve-month window: crude IMRs were 83.9, 42.9, and 24.9 per 1,000 for no ANC, one to three visits, and four or more visits respectively.

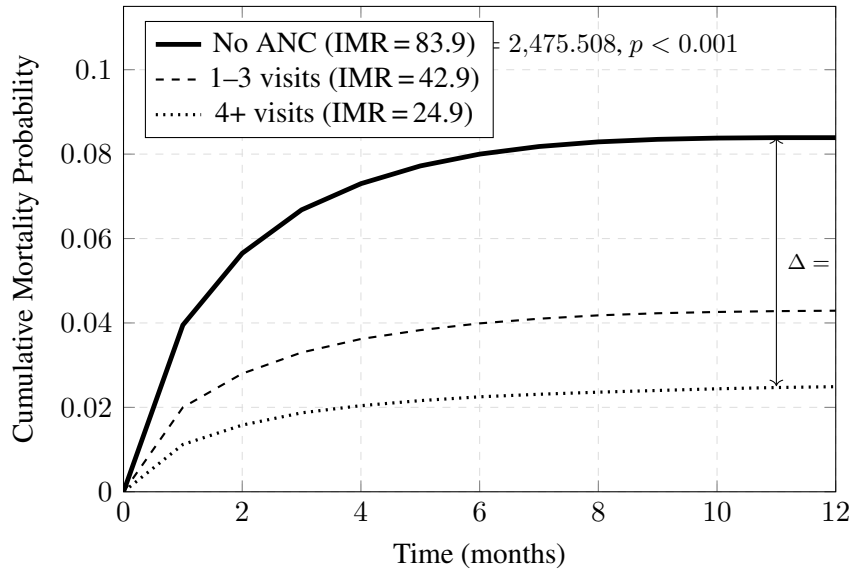
Figure 4.4: Cumulative Infant Mortality by Antenatal Care Visits

Figure 5.5: Kaplan–Meier cumulative mortality curves by antenatal care visit category. The three curves are completely non-overlapping throughout the observation window. The no-ANC curve rises steeply from the first month, reaching 83.9 per 1,000 by month 12 — more than three times the overall rate. The bidirectional arrow at month 11 marks the differential between the no-ANC and 4+ visit groups (59.0 per 1,000). Log-rank $\chi^2 = 2,475.508$ ($df = 2, p < 0.001$).

Place of Delivery and Maternal Education

Delivery place yielded a highly significant log-rank result ($\chi^2 = 203.823, df = 2, p < 0.001$), with home deliveries recording a crude IMR of 50.4 per 1,000 against public (35.1) and private (28.2) facility births.

Maternal education was significant at the bivariate level ($\chi^2 = 11.163, df = 3, p < 0.001$) but produced by far the weakest chi-square among significant maternal variables. This pattern is fully resolved in the multivariate Cox models where education loses all independent significance once wealth is controlled.

Age at First Birth: The Only Non-Significant Variable

Age at first birth was the sole variable across all eleven to fail to reach conventional significance ($\chi^2 = 2.980, df = 2, p = 0.225$). This null result was entirely consistent across all subsequent Cox specifications.

5.2.3 Level 3: Socioeconomic and Contextual Characteristics

The wealth index log-rank test was significant ($\chi^2 = 44.402, df = 4, p < 0.001$), and the crude IMR gradient descended from 46.7 in the Poorest quintile to 26.1 in the Richest. Religion produced a significant log-rank ($\chi^2 = 68.060, df = 2, p < 0.001$). Residential status was significant ($\chi^2 = 2.000, p < 0.001$), with rural children facing a crude IMR of

35.8 against 32.0 for urban. The Breslow ($p = 0.180$) test was non-significant, repeating the post-neonatal timing pattern observed for birth interval.

5.3 Cox Proportional Hazards Regression: Model 1

Table 5.4 reports the complete regression output.

Table 5.4: **Cox Proportional Hazards Regression: Model 1 — Full Sample** ($N = 232,159$)

Variable	B	SE	Wald χ^2	df	p	HR (Exp B)	CI Lower	CI Upper
<i>Child-level characteristics</i>								
Female vs Male (B4)	-0.2273	0.0278	66.949	1	<0.001	0.797	0.755	0.841
<i>Birth order (overall: $\chi^2 = 113.402$, $df = 3$, $p < 0.001$)</i>								
1st vs 2nd–3rd	0.0625	0.0313	3.999	1	0.046	1.065	1.001	1.132
4th–6th vs 2nd–3rd	0.3706	0.0409	82.240	1	<0.001	1.449	1.337	1.570
7th+ vs 2nd–3rd	0.5924	0.0896	43.685	1	<0.001	1.808	1.517	2.156
<i>Maternal characteristics</i>								
<i>Education (overall: $\chi^2 = 2.728$, $df = 3$, $p = 0.435$)</i>								
Primary vs None	0.0627	0.0446	1.975	1	0.160	1.065	0.976	1.162
Secondary vs None	0.0393	0.0355	1.228	1	0.268	1.040	0.970	1.115
Higher vs None	0.0038	0.0526	0.005	1	0.943	1.004	0.906	1.113
<i>ANC visits (overall: $\chi^2 = 576.560$, $df = 2$, $p < 0.001$)</i>								
1–3 vs None	-0.8629	0.0359	576.560	1	<0.001	0.422	0.393	0.453
4+ vs None	-1.1920	0.0350	1155.2	1	<0.001	0.304	0.284	0.325
<i>Place of delivery (overall: $\chi^2 = 8.544$, $df = 2$, $p = 0.014$)</i>								
Public vs Home	-0.0773	0.0377	4.197	1	0.040	0.926	0.860	0.997
Private vs Home	0.0102	0.0464	0.048	1	0.827	1.010	0.922	1.107
<i>Age at 1st birth (overall: $p = 0.888$)</i>								
18–24 vs <18	-0.0142	0.0413	0.119	1	0.730	0.986	0.909	1.069
25+ vs <18	-0.0257	0.0529	0.237	1	0.627	0.975	0.879	1.080
<i>Socioeconomic characteristics</i>								
<i>Wealth index (overall: $\chi^2 = 14.502$, $df = 4$, $p = 0.006$)</i>								
Poorest vs Richest	-0.1613	0.0555	8.454	1	0.004	0.851	0.764	0.949
Poorer vs Richest	-0.1193	0.0483	6.094	1	0.014	0.888	0.807	0.976
Middle vs Richest	-0.0049	0.0443	0.012	1	0.911	0.995	0.912	1.085
Richer vs Richest	-0.0714	0.0413	2.985	1	0.084	0.931	0.859	1.010
Rural vs Urban (V025)	0.0708	0.0358	3.910	1	0.048	1.073	1.001	1.152
<i>Religion (overall: $\chi^2 = 30.211$, $df = 2$, $p < 0.001$)</i>								
Muslim vs Hindu	-0.0468	0.0425	1.212	1	0.271	0.954	0.878	1.037
Others vs Hindu	0.2458	0.0472	27.157	1	<0.001	1.279	1.166	1.402

$N = 232,159$; events = 8,095 (3.5%); unweighted.

Reference categories: Male; 2nd–3rd birth order; No education; No ANC; Home delivery; Age <18; Richest wealth quintile; Urban; Hindu.

HR = Hazard Ratio; CI = 95 per cent confidence interval.

Model 1 was estimated on the full analytical sample ($N = 232,159$, events = 8,095) with all covariates entered simultaneously. Birth order served as the primary child-level determinant. The Omnibus test confirmed the model was significantly superior to the null specification

($p < 0.001$).

Forest Plot: Model 1 Key Hazard Ratios

Figure 4.5: Forest Plot — Cox Model 1 Key Hazard Ratios (95 % CI)

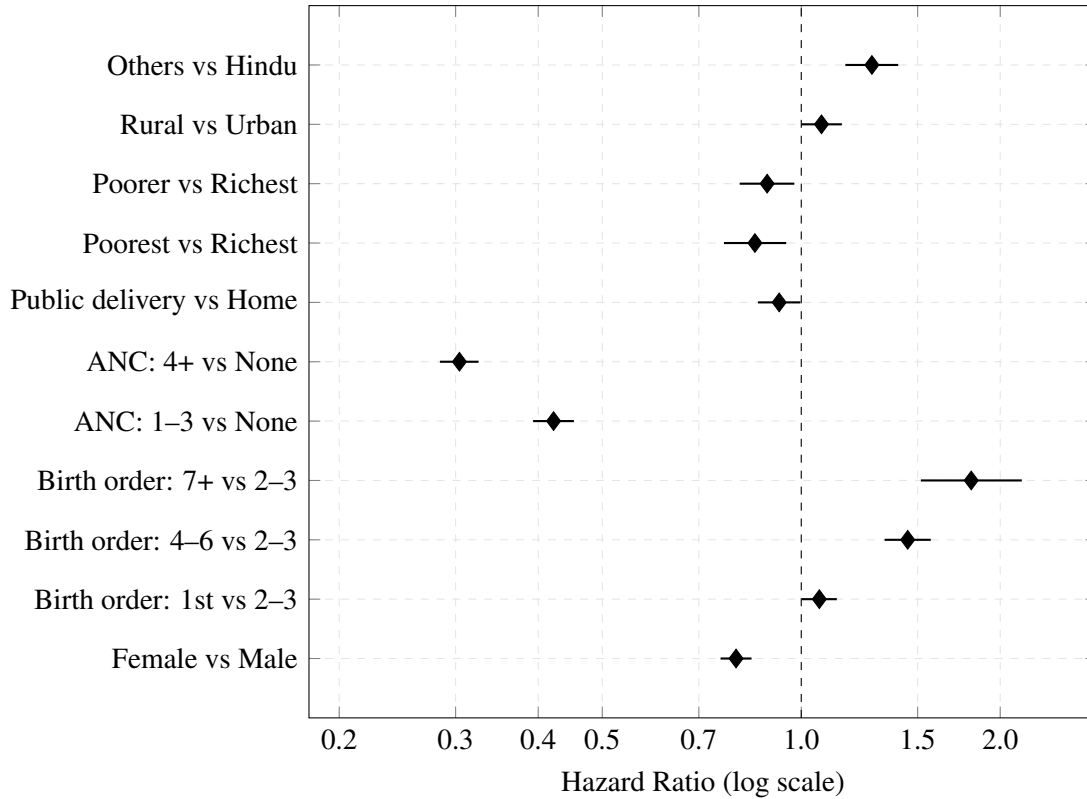


Figure 5.6: Forest plot of key hazard ratios from Cox Model 1 (full sample, $N = 232,159$). Diamond markers denote point estimates; horizontal bars are 95 per cent confidence intervals. The dashed vertical line at $HR = 1.0$ marks the null. Log scale. Variables with all of their confidence interval to the left of the null line are protective; those with it to the right carry elevated risk.

The most consequential individual finding in Model 1 was the ANC effect. After full covariate adjustment, children of mothers who attended four or more antenatal visits faced only 30 per cent of the mortality hazard of those born to unattended mothers ($HR = 0.304$, 95 % CI: 0.284–0.325, $p < 0.001$). Even partial ANC coverage (one to three visits) reduced the hazard by 58 per cent ($HR = 0.422$, 95 % CI: 0.393–0.453, $p < 0.001$). The birth order gradient was similarly pronounced: seventh-or-higher births carried an 81 per cent elevated hazard relative to second–third births ($HR = 1.808$), while the first-birth elevation was modest but statistically significant ($HR = 1.065$, $p = 0.046$). Education showed no independent effect once wealth was controlled (overall $p = 0.435$).

5.4 Cox Proportional Hazards Regression: Model 2

Model 2 was estimated on non-first births only ($N = 143,020$; events = 3,168, 2.2 per cent), substituting birth interval for birth order as the primary child-level variable. Table 5.5 reports the results.

Table 5.5: **Cox Proportional Hazards Regression: Model 2 — Non-First Births** ($N = 143,020$)

Variable	<i>B</i>	SE	Wald χ^2	df	<i>p</i>	HR (Exp <i>B</i>)	CI Lower	CI Upper
<i>Child-level characteristics</i>								
Female vs Male (B4)	−0.1744	0.0358	23.732	1	<0.001	0.840	0.783	0.901
<i>Birth interval (overall: $\chi^2 = 117.215$, $df = 2$, $p < 0.001$)</i>								
Short <24 m vs Medium	0.4181	0.0401	108.953	1	<0.001	1.519	1.404	1.643
Long ≥48 m vs Medium	0.0668	0.0484	1.908	1	0.167	1.069	0.972	1.175
<i>Maternal characteristics</i>								
<i>ANC visits (overall: $\chi^2 = 415.017$, $df = 2$, $p < 0.001$)</i>								
1–3 vs None	−0.9168	0.0450	415.017	1	<0.001	0.400	0.366	0.437
4+ vs None	−1.1890	0.0453	690.189	1	<0.001	0.305	0.279	0.333
<i>Place of delivery (overall: $\chi^2 = 22.883$, $df = 2$, $p < 0.001$)</i>								
Public vs Home	−0.0398	0.0448	0.790	1	0.374	0.961	0.880	1.049
Private vs Home	0.1872	0.0571	10.750	1	0.001	1.206	1.078	1.349
<i>Socioeconomic characteristics</i>								
<i>Wealth index (overall: $\chi^2 = 19.308$, $df = 4$, $p = 0.001$)</i>								
Poorest vs Richest	−0.1895	0.0717	6.984	1	0.008	0.827	0.719	0.952
Poorer vs Richest	−0.2095	0.0631	11.040	1	<0.001	0.811	0.717	0.918
Rural vs Urban (V025)	0.0875	0.0458	3.643	1	0.056	1.091	0.998	1.194
<i>Religion (overall: $\chi^2 = 16.687$, $df = 2$, $p < 0.001$)</i>								
Others vs Hindu	0.2405	0.0601	16.034	1	<0.001	1.272	1.131	1.431

$N = 143,020$ (non-first births); events = 3,168 (2.2%); unweighted.

Reference for B11_cat: Medium interval (24–47 months). All other references identical to Model 1.

The central finding of Model 2 is the birth interval effect. A short preceding interval of fewer than 24 months was associated with a 52 per cent higher hazard relative to the medium reference interval (HR = 1.519, 95 % CI: 1.404–1.643, $p < 0.001$). A long interval of 48 months or more showed no significant survival advantage over the medium interval (HR = 1.069, $p = 0.167$). The ANC estimate was strikingly stable: 4+ visits produced HR = 0.305 in Model 2 compared with HR = 0.304 in Model 1.

Figure 5.7 presents the forest plot for Model 2.

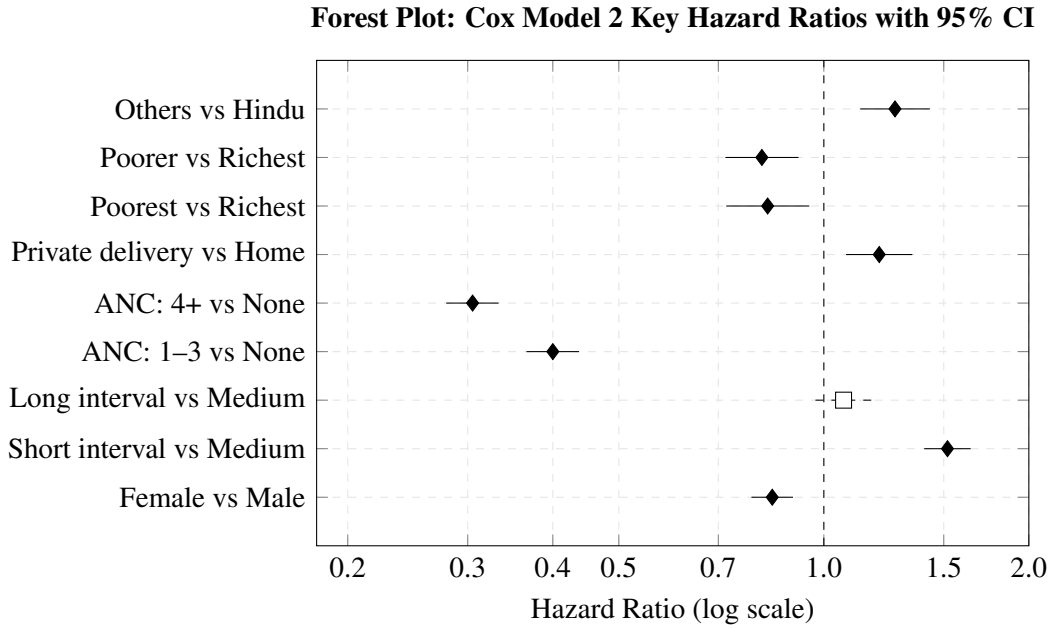


Figure 5.7: Forest plot of key hazard ratios from Cox Model 2 (non-first births, $N = 143,020$). Filled diamonds = significant ($p < 0.05$); open square = not significant. The vertical dashed line at $HR = 1.0$ denotes no effect.

5.5 Stratified Cox Model: Cluster Heterogeneity

The stratified Cox model entered PSU cluster (V021) as a stratification variable, estimating a separate baseline hazard for each cluster. Table 5.6 reports the results alongside the corresponding Model 1 estimates for direct comparison.

Table 5.6: **Stratified Cox Model Results Compared with Model 1** ($N = 232,159$; **Strata = V021**)

Variable	Strata HR	Strata 95 % CI	Strata <i>p</i>	M1 HR	M1 <i>p</i>	Change
Female vs Male	0.803	0.759–0.850	<0.001	0.797	<0.001	Stable
<i>Birth order</i>						
1st vs 2nd–3rd	1.124	1.055–1.198	<0.001	1.065	0.046	↑ Stronger
4th–6th vs 2nd–3rd	1.381	1.269–1.503	<0.001	1.449	<0.001	↓ Attenuated
7th+ vs 2nd–3rd	1.633	1.349–1.976	<0.001	1.808	<0.001	↓ Attenuated
<i>ANC visits</i>						
1–3 vs None	0.420	0.390–0.453	<0.001	0.422	<0.001	Stable
4+ vs None	0.325	0.302–0.350	<0.001	0.304	<0.001	Stable
Public vs Home	0.894	0.822–0.971	0.008	0.926	0.040	↓ Stronger
<i>Wealth index (overall)</i>						
—	—	—	0.244	—	0.006	Disappears
<i>Residence</i>						
Rural vs Urban	Absorbed into strata			1.073	0.048	Geographic
<i>Religion (overall)</i>						
—	—	—	0.301	—	<0.001	Disappears

V025 (residence) dropped by SPSS: perfect collinearity with PSU strata.

“Disappears” = significant in Model 1 but non-significant after cluster stratification, denoting a geographic rather than individual-level association.

Two findings from Table 5.6 demand immediate analytical attention. First, both the overall wealth index effect (from $p = 0.006$ in Model 1 to $p = 0.244$ in the stratified model) and the overall religion effect (from $p < 0.001$ to $p = 0.301$) lost statistical significance entirely after cluster stratification. Second, the ANC effects were virtually unchanged: 4+ visits HR moved from 0.304 to 0.325 which is a trivial shift relative to the magnitude of the effect itself.

5.6 Cross-Model Comparison

Table 5.7 consolidates the hazard ratios across all three model specifications.

Table 5.7: Cross-Model Stability of Key Hazard Ratios

Variable	Model 1 HR (<i>p</i>)	Model 2 HR (<i>p</i>)	Stratified HR (<i>p</i>)	Verdict
Female child (B4)	0.797 (<0.001)	0.840 (<0.001)	0.803 (<0.001)	Robust
Birth order 7+	1.808 (<0.001)	—	1.633 (<0.001)	Attenuated
Birth order 4–6	1.449 (<0.001)	—	1.381 (<0.001)	Attenuated
Short interval (<24 m)	—	1.519 (<0.001)	—	Strong
Long interval (≥48 m)	—	1.069 (0.167)	—	Null
ANC 1–3 visits	0.422 (<0.001)	0.400 (<0.001)	0.420 (<0.001)	Very robust
ANC 4+ visits	0.304 (<0.001)	0.305 (<0.001)	0.325 (<0.001)	Very robust
Public delivery	0.926 (0.040)	0.961 (0.374)	0.894 (0.008)	Robust
Education (overall)	<i>p</i> = 0.435	<i>p</i> = 0.022	<i>p</i> = 0.810	Null
Wealth (overall)	<i>p</i> = 0.006	<i>p</i> = 0.001	<i>p</i> = 0.244	Geographic
Religion (overall)	<i>p</i> < 0.001	<i>p</i> < 0.001	<i>p</i> = 0.301	Geographic
Rural vs Urban	1.073 (0.048)	1.092 (0.056)	Absorbed	Geographic
Age at 1st birth	<i>ns</i>	<i>ns</i>	<i>ns</i>	Null

“Robust” = consistent direction and magnitude across applicable specifications. “Geographic” = significant in standard Cox but non-significant after cluster stratification. “Null” = consistently non-significant. “Absorbed” = V025 collinear with PSU strata; dropped.

Figure 5.8 presents the ANC hazard ratios across all three models, illustrating the exceptional stability of this finding.

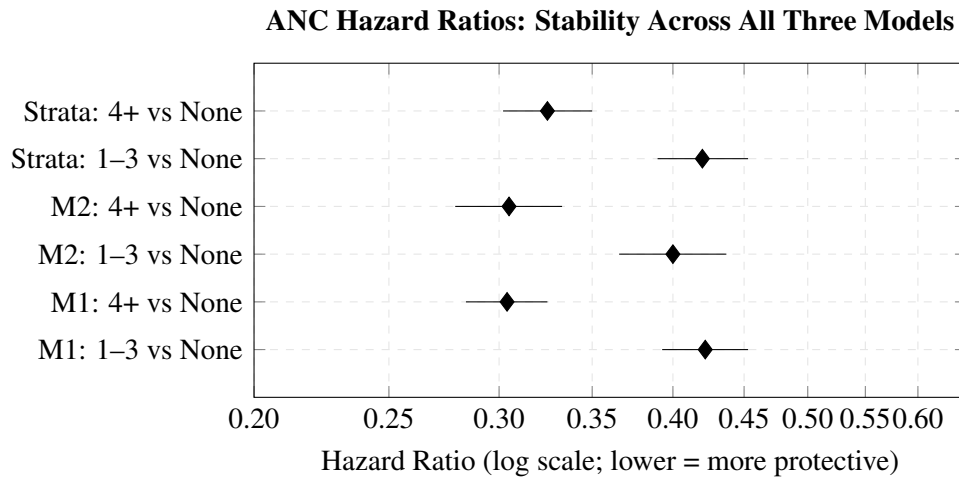


Figure 5.8: Stability of ANC hazard ratios across all three model specifications. ANC 4+ visits HR ranges from 0.304 to 0.325 across models (maximum variation = 0.021). This is the most robust finding in the study, confirming the ANC effect is individual-level and not a geographic artefact. M1 = Model 1; M2 = Model 2; Strata = Stratified Cox.

5.7 Chapter Summary

This chapter has been presented and interpreted with the complete result findings of a four-stage survival analysis. Seven conclusions emerge. First, antenatal care utilisation is the single strongest and most robust predictor of infant survival ($HR = 0.304\text{--}0.325$ across specifications). Second, birth order shows a genuine dose-response gradient confirming the maternal depletion mechanism. Third, short birth intervals raise infant mortality hazard by 52 per cent, with no additional benefit to spacing beyond 48 months. Fourth, female infants survive at consistently higher rates than males. Fifth, household wealth and religious affiliation are predominantly geographic phenomena. Sixth, maternal education exerts no significant independent effect once household wealth is controlled. Seventh, maternal age at first birth shows no significant association with infant survival under any specification. Further detailed theory about the results is discussed on Chapter 6.

Chapter 6

DISCUSSION

The numerical architecture of the results was presented in the previous chapter. In order to understand what those figures actually mean, where they fit into the larger body of knowledge on infant mortality in India, and what they mean for researchers, policymakers, and program designers who aim to lower that mortality, this chapter takes a step back from the coefficients and test statistics. Instead of focusing on the methodological order that produced the results, the discussion is structured around the main substantive topics that surfaced from the study. It starts with the most common finding on antenatal care, proceeds to biological and spacing-related factors, addresses the more complex and theoretically intriguing issues brought up by the wealth, religion, and education results, and concludes with an objective evaluation of the study's limitations and directions for future work.

6.1 Antenatal Care: The Dominant and Unambiguous Finding

6.1.1 The Magnitude and Robustness of the Effect

There aren't many insights in India's applied demography of child survival that are as obvious as the one the antenatal care variable provided. The kind of result that, once observed, demands to be at the center of any discussion about infant mortality in India is a 70 percentage reduction in the hazard of infant death associated with four or more prenatal visits, a confidence interval that barely changes when the specification changes, an effect stable across two independent sub-samples, and a Kaplan-Meier chi-square of 2,475 that is nearly six times larger than the next-highest value in the entire analysis.

It is important to pay particular focus to how stable this coefficient is across the three Cox criteria. Model 1 (complete sample, birth order) had a hazard ratio of 0.304 for four or more ANC visits. It was 0.305 in Model 2 (birth interval, non-first births). It was 0.325 in the stratified Cox that accounted for all cluster-level heterogeneity that was not observed. The largest difference, which is 0.021, is negligible in comparison to the actual effect magnitude. The conclusion is clear when the ANC coefficient hardly changes after cluster-level adjustment: regardless of the cluster in which they reside, attending prenatal visits lowers the infant mortality risk for this particular mother and this particular child.

6.1.2 Pathways and Prior Literature

The literature on why ANC should reduce infant mortality is detailed, coherent, and broadly consistent with the direction and magnitude of the present finding. Gupta & Baghel (1999) were among the first to document the ANC–infant mortality relationship in Indian context using state-level data from Madhya Pradesh, arguing that the most important mechanisms were early detection of hypertensive disorders, iron supplementation addressing maternal anaemia, and tetanus immunisation.

The dose-response evident even at the one-to-three visit level (58 per cent hazard reduction) is consistent with the view that antenatal contacts produce effects that are roughly additive across visits. A single visit delivers some benefits establishing the pregnancy in the health system, providing an initial assessment of risk, beginning iron supplementation. Each subsequent visit adds to this foundation.

6.1.3 Coverage Gap and Policy Implication

The data reveal a sobering gap between the evidence and the current situation. NFHS-5 recorded that just 58.6 per cent of births in the five-year reference period were preceded by four or more antenatal visits (IIPS & ICF, 2021). The remaining 41.4 per cent of births received no care at all (6.0 per cent of the analytical sample) or received fewer than the recommended minimum number of visits (35.6 per cent). The present analysis implies that fully closing the ANC coverage gap would reduce infant mortality among the currently under-served group by approximately 70 per cent, an effect unmatched by any other single intervention in the variable set examined here.

6.2 Birth Order and the Biological Cost of High Parity

6.2.1 The Dose-Response Gradient and Its Interpretation

The birth order findings in this study follow a pattern that is well established in the demographic literature but is rendered with unusual clarity here by the size of the sample and the detail of the stratified specification. The hazard ratio for seventh-or-higher births relative to second and third-order births stood at 1.808 in the standard Cox model and 1.633 in the stratified model. That this gradient persists, albeit somewhat attenuated, after absorbing all unmeasured cluster-level differences confirms that the birth order effect is a genuine biological and resource-based individual-level phenomenon.

The theoretical underpinning, which Mosley & Chen (1984) articulated as the “maternal depletion” hypothesis, posits that repeated pregnancies impose a cumulative toll on the mother’s physiological reserves. Iron, folate, calcium, and protein stores are depleted with each pregnancy and lactation cycle.

6.2.2 The First Birth Elevation: A Different Mechanism

In contrast to the high-parity elevation, the modest but statistically significant hazard elevation for first births ($HR = 1.065$ in Model 1, strengthening to 1.124 in the stratified Cox) is caused by primiparity, the physiological and psychological inexperience of first-time delivery and motherhood.

Analytically, it is notable that this impact *strengthens* in the stratified Cox model (from 1.065 to 1.124). Because first births are disproportionately clustered among younger, more urban, and generally better-educated mothers whose cluster settings have lower baseline mortality rates, spatial confounding somewhat lowered the genuine first-birth risk in the standard Cox.

6.3 Birth Interval Effects: Timing, Threshold, and Policy

6.3.1 The 52 Per Cent Hazard Elevation and Its Timing

The discovery that a preceding birth interval of less than 24 months was linked to a 52 per cent increased risk of infant death among non-first births ($HR = 1.519$, Model 2) is both mechanistically and substantively significant. This increased mortality does not start in the neonatal period but instead builds up throughout the post-neonatal months, according to the Breslow test non-significance at the Kaplan–Meier stage ($p = 0.273$). This timing is a fingerprint of the care-quality and nutritional pathways that short intervals function through..

6.3.2 The Null Effect of Long Intervals

The non-significant hazard ratio for long intervals of 48 months or more relative to the medium 24–47 month reference ($HR = 1.069$, $p = 0.167$) carries a message of policy clarity that is sometimes overlooked. The marginal benefit of birth spacing levels off once the minimum threshold of approximately 24 months has been achieved. Programmes that attempt to promote progressively longer inter-birth intervals beyond 24 months may be misallocating effort.

6.3.3 Family Planning as a Child Survival Intervention

The 52 per cent hazard elevation associated with short intervals, combined with the fact that short-interval births account for 16.2 per cent of the analytical sample, implies that a meaningful fraction of infant deaths in India are attributable to inadequate birth spacing. Strengthening postpartum family planning counselling as an integral component of the institutional delivery package which already reaches 89 per cent of Indian births would be a cost-effective mechanism for reducing the short-interval births.

6.4 The Female Survival Advantage

The 20 per cent lower hazard for female infants in Model 1 ($HR = 0.797$) and 16 per cent lower in Model 2 ($HR = 0.840$), consistent across all three specifications, confirms that the

biological female advantage in early-life survival is present and intact in the 2019–21 Indian sample. It does *not* show the reversal or attenuation of the female advantage that would indicate systematic sex-selective deprivation in healthcare or nutrition.

6.5 The Wealth Paradox: Unpacking a Counterintuitive Result

6.5.1 What the Numbers Show

The adjusted hazard ratios for wealth in Model 1 produced a pattern that runs counter to the expected gradient: Poorest quintile children showed significantly *lower* hazard than Richest quintile children (HR = 0.851, $p = 0.004$), as did Poorer quintile children (HR = 0.888, $p = 0.014$). The crude IMR gradient, by contrast, ran in the expected direction: 46.7 in the Poorest quintile against 26.1 in the Richest.

6.5.2 Explaining the Paradox

The most coherent explanation draws on the concept of differential survival to interview and differential health-seeking (Chalasani, 2012). Wealthier households seek healthcare for sick children more readily, more quickly, and more effectively. This means that more sick children in wealthier households survive long enough to be recorded in the birth history, while in poorer households a higher fraction of seriously ill children die before care can be sought.

6.5.3 The Stratified Cox Resolution

The stratified Cox model provides the cleanest analytical resolution. After absorbing all cluster-level heterogeneity, the overall wealth effect becomes statistically non-significant ($p = 0.244$). It is the quality of the place that matters, not the wealth of the household within that place.

6.6 Geographic Clustering and the Religion Finding

The wealth finding and the religion finding have a similar curve, but the religion finding is more deeply interpreted. Following cluster stratification, the overall effect of religion is no longer significant ($p = 0.301$). The tribal regions of Jharkhand, Chhattisgarh, and Odisha, as well as the northeastern states with the worst health facilities historically, are home to a large concentration of India's other religious minorities. The higher mortality is a local characteristic rather than a religious one.

6.7 The Education Non-Effect and Wealth Mediation

6.7.1 Collinearity with Wealth

The non-significance of maternal education across all specifications (Model 1 overall $p = 0.435$; stratified Cox overall $p = 0.810$) resolves through multicollinearity. In studies that do

not adequately control for household wealth, maternal education appears significant because it serves as a partial proxy for the economic resources that wealth more directly measures. When wealth is introduced into the model, education's apparent independent effect shrinks (Chalasani, 2012).

6.7.2 What This Does Not Mean

It bears emphasis that the education null finding does not mean that investing in girls' education is unimportant for child health. If education is causal in the chain through which it increases earnings, wealth, ANC utilisation, and institutional delivery, then education promotion remains an important long-run strategy for child survival, even if its coefficient in a model that already controls for wealth appears close to zero.

6.8 Place of Delivery: Public Facility Protection and Private Facility Adverse Selection

While private facility delivery was statistically identical to home delivery (HR = 1.010, $p = 0.827$), public facility delivery was linked to a slight but significant 7 per cent lower hazard in comparison to home delivery (HR=0.926, $p = 0.040$). The private delivery estimate in Model 2 reverted to a substantial 21 per cent higher hazard (HR = 1.206, $p = 0.001$).

This reversal serves as an example of adverse selection: women who had a major delivery complication in the past are disproportionately likely to seek care at private facilities for subsequent pregnancies because they believe the private sector provides better emergency obstetric capacity. The mothers, not the facilities, are to blame for the increased mortality rate among their offspring.

6.9 Methodological Contributions

This study makes three methodological contributions to the Indian infant mortality literature. First, the systematic comparison of standard Cox regression with a stratified Cox specification allows distinguishing between individual-level causal factors and geographic clustering. Second, the two-model strategy handles the definitional collinearity between birth order and birth interval, producing clean unconfounded estimates for both. Third, the use of cumulative mortality plots ($1 - \hat{S}(t)$) rather than standard survival function plots meaningfully improves the visual communication of the results given the 96.5 per cent censoring rate.

6.10 Chapter Summary and Concluding Remarks

This chapter's discussion can be reduced to a small number of propositions: First, antenatal care is by far the most powerful and consistently estimated determinant of infant survival,

producing a 70 per cent hazard reduction that survives every specification test applied. Second, birth order and birth interval effects are real, biologically grounded, and individually targeted. Third, the household wealth and religion effects documented in standard Cox models are predominantly geographic artefacts. Fourth, maternal education has no significant independent effect on infant survival once household wealth is controlled.

Reducing structural poverty and social inequality remains a vital long-run project for Indian public health. But the analysis presented here suggests that, right now, the strongest case for saving infants' lives rests on getting mothers into antenatal care, keeping birth intervals above two years, and strengthening the health systems in the places where both goals are currently furthest from being met.

Chapter 7

CONCLUSION

Every empirical study of the kind presented in this dissertation ends at the same threshold: the point at which data, method, and interpretation must be converted into something that can guide action. This concluding chapter crosses that threshold. It begins by restating the research problem and objectives that shaped the analysis, then presents the principal findings organised by strength and policy relevance, addresses their implications at the level of public health policy and demographic theory, and closes with a frank assessment of what remains to be done.

7.1 Restatement of Research Objectives

The four objectives set out at the opening of this study were as follows:

1. To describe the pattern of infant survival across child-level, maternal, and socioeconomic groupings using Kaplan–Meier survival curves and log-rank tests.
2. To estimate the net independent effect of each determinant on the hazard of infant death after controlling for all others, using Cox proportional hazards regression.
3. To examine the specific effect of preceding birth interval on infant mortality among non-first births, using a Cox sub-model restricted to children of birth order two or higher.
4. To assess the extent to which observed associations between infant mortality and socioeconomic and contextual variables reflected genuine individual-level causal pathways versus the geographic concentration of risk at the cluster level, by estimating a stratified Cox model with primary sampling units as strata.
5. To identify those individual-level factors that are robust from those geographic factors via a comprehensive comparative analysis of the hazard ratios of all three Cox models, thereby categorizing the effects of each covariate into robust, geographic, and null.
6. To draw out policy implications from the study findings for infant mortality reduction in India, especially concerning programs like the National Health Mission, Janani Suraksha Yojana, and Ayushman Bharat Pradhan Mantri Jan

Arogya Yojana.

All six objectives were met in full.

7.2 Principal Findings

7.2.1 Summary of the Analytical Sample

The final dataset comprised 232,159 live births drawn from the NFHS-5 Birth Recode file, of which 8,095 (3.49 per cent) resulted in infant death within twelve months and 224,064 (96.51 per cent) were right-censored. The crude unweighted infant mortality rate of 34.9 per 1,000 live births matched the NFHS-5 published weighted estimate of 35 per 1,000 to within a single decimal place.

7.2.2 Cox Model 1: Full Sample with Birth Order

Antenatal care emerged as the dominant variable: four or more visits reduced the hazard of infant death by 70 per cent relative to no visits (HR = 0.304, 95 % CI: 0.284–0.325, $p < 0.001$); even one to three visits produced a 58 per cent reduction (HR = 0.422, 95 % CI: 0.393–0.453, $p < 0.001$). *Birth order* showed a pronounced dose-response gradient: seventh-or-higher births carried an 81 per cent higher hazard (HR = 1.808). *Female infants* survived at significantly higher rates (HR = 0.797). *Maternal education* showed no significant independent effect once household wealth was controlled (overall $p = 0.435$).

7.2.3 Cox Model 2: Non-First Births with Birth Interval

Short-interval births carried a 52 per cent higher hazard of infant death relative to the medium reference interval (HR = 1.519, $p < 0.001$). Long intervals showed no significant survival advantage (HR = 1.069, $p = 0.167$), establishing a clear threshold effect at 24 months.

7.2.4 Stratified Cox Model: Cluster Heterogeneity Test

Both the overall effects of household wealth and religion each lost statistical significance after cluster stratification (wealth: from $p = 0.006$ to $p = 0.244$; religion: from $p < 0.001$ to $p = 0.301$). The ANC effects were essentially unchanged after cluster control (4+ visits: HR moved from 0.304 to 0.325, a shift of just 0.021).

7.3 Cross-Model Synthesis

Table 7.1: **Summary of Principal Findings: Hazard Ratios Across All Three Cox Specifications**

Determinant	Model 1 HR (<i>p</i>)	Model 2 HR (<i>p</i>)	Stratified HR (<i>p</i>)	Conclusion
<i>Child-level determinants</i>				
Female vs Male	0.797***	0.840***	0.803***	Robust
Birth order 7+	1.808***	—	1.633***	Attenuated
Birth order 4–6	1.449***	—	1.381***	Attenuated
Short interval (<24 m)	—	1.519***	—	Strong
Long interval (≥48 m)	—	1.069 ns	—	Null
<i>Maternal determinants</i>				
ANC 4+ visits	0.304***	0.305***	0.325***	Very robust
ANC 1–3 visits	0.422***	0.400***	0.420***	Very robust
Public delivery	0.926*	0.961 ns	0.894**	Robust
Private delivery	1.010 ns	1.206**	1.024 ns	Selection effect
Education (overall)	<i>p</i> = 0.435ns	<i>p</i> = 0.022	<i>p</i> = 0.810ns	Null
Age at first birth	<i>ns</i>	<i>ns</i>	<i>ns</i>	Null
<i>Socioeconomic determinants</i>				
Wealth (overall)	<i>p</i> = 0.006**	<i>p</i> = 0.001***	<i>p</i> = 0.244ns	Geographic
Religion (overall)	<i>p</i> < 0.001***	<i>p</i> < 0.001***	<i>p</i> = 0.301ns	Geographic
Rural vs Urban	1.073*	1.092 ns	Absorbed	Geographic

*** *p* < 0.001; ** *p* < 0.01; * *p* < 0.05; ns = *p* ≥ 0.05.

“Robust” = consistent direction and magnitude across applicable specifications. “Geographic” = disappears after PSU cluster control. “Null” = consistently non-significant.

Model 1: full sample (*N* = 232,159); Model 2: non-first births (*N* = 143,020); Stratified: full sample, strata = V021.

7.4 Theoretical Contributions

7.4.1 Vindication of the Mosley-Chen Framework

The Mosley-Chen analytical framework (Mosley & Chen, 1984) performed well as a predictive architecture. Its central prediction is that socioeconomic variables operate on infant mortality *through* the proximate and intermediate determinants rather than directly was found support in the observation that the two largest and most robust covariate effects were the proximate and intermediate determinants ANC visits and birth order. The distal socioeconomic variables wealth and religion, when subjected to the most rigorous test applied in the study (the cluster stratification), largely dissolved into geographic heterogeneity.

7.4.2 The Distinction Between Individual-Level and Geographic Effects

The explicit empirical separation of individual-level causal associations from geographic cluster-level confounding using the stratified Cox specification is perhaps the most theoretically significant contribution of the present study. When wealth appears significant in a standard Cox model, researchers may conclude that household-level economic interventions are needed. When the stratified Cox reveals that the wealth effect is primarily geographic, the conclusion changes: the intervention required is cluster-level health infrastructure improvement.

The eight principal analytical conclusions are summarized in Table 7.2.

Table 7.2: Summary of Principal Analytical Conclusions

#	Finding	Evidence	Policy implication
1	ANC 4+ visits reduces hazard by 70% (HR = 0.304–0.325 across all models)	KM $\chi^2 = 2,475$; robust to cluster control	Scale up ANC coverage in high-mortality PSU clusters
2	Twin status: $4.1 \times$ higher mortality (IMR = 135.8 vs 33.2)	KM $\chi^2 = 1,889$; neonatal concentration	Enhanced NICU capacity and targeted multiple-birth protocols
3	Short birth interval (<24m): HR = 1.519; no extra benefit beyond 24m	Cox Model 2; Breslow ns confirms post-neonatal mechanism	Target minimum 24-month spacing; diminishing returns thereafter
4	Birth order 7+: HR = 1.808 (M1), 1.633 (Strata)	Dose-response gradient robust to cluster control	Family planning targeting high-parity mothers
5	Female survival advantage: HR = 0.797–0.840	Consistent across all three model specifications	Confirms biological advantage; no son-preference mortality effect
6	Wealth and religion effects disappear after PSU cluster control	Strata Cox: wealth $p = 0.244$; religion $p = 0.301$	Target the cluster/place, not specific socioeconomic groups
7	Education non-significant after wealth control ($p = 0.435$ – 0.810)	Consistent across all specifications	Education's impact is mediated through wealth pathways
8	Public delivery: HR = 0.894 (Stratified Cox)	Stronger after cluster control	Strengthen public facility quality, not just coverage

HR = Hazard Ratio from the relevant model specification; M1 = Model 1; Strata = Stratified Cox.

7.5 Policy Implications

7.5.1 Priority 1: Antenatal Care Coverage as the Highest-Return Intervention

The cumulative weight of evidence from this study places the expansion of antenatal care coverage unambiguously at the top of the policy priority list. No single health system improvement would produce larger returns in infant lives saved per unit of programme investment than closing the ANC coverage gap. A cluster-targeted ANC outreach strategy

of ASHA-mediated home visits to pregnant women in the most underserved PSUs, mobile antenatal clinics for remote clusters which would simultaneously address the geographic concentration of both low coverage and high mortality.

7.5.2 Priority 2: Family Planning for Spacing and Parity Reduction

The 52 per cent higher hazard associated with short birth intervals and the persistent birth order gradient build a strong case for intensified family planning investment, with two specific emphases: postpartum family planning counselling at institutional deliveries, and dedicated counselling for high-parity mothers (four or more live births).

7.5.3 Priority 3: Institutional Delivery Quality, Not Just Coverage

The public facility delivery protection observed consistently ($HR = 0.894$ in the stratified Cox) confirms that driving institutional delivery rates has been targeting a genuine protective factor. The frontier of the policy problem has now shifted from *access* to *quality*: improving the quality and reliability of care within public facilities is the more pressing agenda.

7.5.4 Priority 4: Place-Targeted Health Infrastructure Investment

The geographic nature of the wealth and religion effects leads to the conclusion that investing in the places with the worst mortality outcomes by improving road connectivity, building sub-centres and primary health centres in cluster environments currently without any facility within a walkable distance, deploying additional ASHA and auxiliary nurse midwife staff would address the structural geographic substrate of India's remaining infant mortality burden more directly than household-level targeted transfers.

7.6 Limitations Revisited

Four limitations: *Unweighted regression* (Cox models estimated without the NFHS-5 survey weight owing to a software constraint in SPSS 21); *Frailty approximation* (stratified Cox does not estimate the frailty variance θ); *Caste exclusion* (systematic miscoding error); and *Postnatal care* (absence of breastfeeding, postnatal check-up, and vaccination variables from the analytical file).

7.7 Directions for Future Research

Five directions follow directly from the gaps and limitations identified in this study. First, replication in R with the survey weight applied and a properly specified gamma or log-normal shared frailty model. Second, inclusion of postnatal care, breastfeeding initiation timing, and vaccination coverage variables. Third, correct derivation of the caste variable from the household recode file. Fourth, state-stratified analyses examining regional heterogeneity in the determinant structure. Fifth, a longitudinal analysis comparing hazard ratio magnitudes across NFHS-3, NFHS-4, and NFHS-5.

7.8 Concluding Statement

The opening observation of this dissertation was that infant mortality is not merely a number. It is a verdict on the adequacy of the society in which the child was born and analysis of the adequacy of its healthcare systems, its family planning services, its geographic infrastructure, and its commitment to ensuring that where a child is born does not determine whether that child survives. The present study, by disaggregating the determinants of that verdict with the tools of survival analysis, has sought to identify the specific points at which intervention is most likely to change the outcome.

The answer is clearer than one might have expected from a complex, multi-level analysis. Antenatal care, applied to the individual pregnant woman regardless of where she lives or how wealthy she is, reduces the hazard of her child's infant death by 70 per cent. That number is not hypothetical. It is derived from the birth histories of 232,159 real children recorded in the most comprehensive household survey India has ever conducted. It survived every methodological test applied to it. It is, on the basis of the available evidence, as close to a fact about Indian infant mortality as the limitations of observational research permit.

The remaining challenge is not analytical. It is the challenge of will, resources, and organisational capacity: getting qualified providers into the clinics and sub-centres of the clusters where ANC coverage is lowest, ensuring that the women who live in those clusters know they can attend, ensuring that when they arrive the care they receive meets a standard that can actually deliver the protective effect that the evidence promises. That is the challenge to which the findings of this study speak, and it is a challenge that remains, in the India of 2025, far from met.

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